A parameter-sparse state-space framework for characterizing coastal dune-aquifer architecture from manual dipwell records

M. Hollingham

Draft — 2026

Abstract

Study region: Newborough Warren, Anglesey, Wales — a coastal dune system on the Irish Sea monitored by a manual dipwell network of 88 wells over 21 years alongside a single climate station.

Study focus: Dune-slack vegetation occupies a band of water-table depth only a few tens of centimetres wide, yet characterizing the aquifer beneath it is difficult in unconsolidated sand, where distributed flow models demand hydraulic-conductivity and aquifer-thickness fields that are expensive to measure and poorly constrained. We ask whether the routine dipwell record alone can resolve dune-aquifer architecture, and present a parameter-sparse framework that does so without coring, hydraulic testing or calibrated flow modelling: hierarchical clustering of hydrograph dynamics coupled to a three-term state-space model — recharge, atmospheric draw and head-dependent drainage referenced to a fixed sub-surface displacement datum — fitted at each well from monthly rainfall and temperature.

New hydrological insights for the region: Five hydrogeological clusters emerge, independently confirmed by their mechanistic coefficients. Interception-corrected specific yields converge across them, showing the behavioural contrasts to be surface- and land-cover-mediated rather than differences in subsurface storage. The displacement formulation closes the monthly water balance to within 1.8% of losses and raises median iterative Nash–Sutcliffe efficiency from −0.03 to 0.72 over a linear benchmark. A coastal-retreat gradient, strongest at the shore and decaying to nothing over an inland reach of about 900 m to 1 km, is separated from the site-wide climate background and explains the anomalous decline of the most coast-proximal cluster.

Keywords: coastal dune aquifer; dune slack; state-space model; specific yield; groundwater characterization; coastal retreat

Highlights

- Dune-aquifer architecture resolved from manual dipwells and one climate station
- State-space model with a displacement datum closes the water balance to within 1.8%
- Iterative forecast skill rises from NSE −0.03 to 0.72 over a linear benchmark
- Interception-corrected specific yields converge across clusters: storage is uniform
- Coastal retreat resolved as a forcing distinct from climate on long-term decline

1. Introduction

Coastal dune slacks are among the most hydrologically sensitive habitats in north-west Europe. The seasonally fluctuating water table that defines them sits within a narrow envelope: the wet- and dry-slack vegetation communities of conservation interest occupy a band of summer-minimum depth only a few tens of centimetres wide (Davy et al., 2006; Curreli et al., 2013). Because that envelope is set by groundwater rather than by surface management, the conservation status of a dune system cannot be understood without a quantitative account of how its aquifer fills, stores and releases water — how recharge enters the water table, how evaporative demand and lateral drainage remove it, and how these processes vary across the geomorphological range of the site.

Characterizing that architecture by conventional hydrogeological means is difficult in dune settings. Distributed groundwater models of the MODFLOW family require hydraulic conductivity fields, aquifer-thickness surfaces and boundary fluxes that are expensive to measure and poorly constrained in unconsolidated, rapidly transmissive dune sand; at Newborough Warren itself, an earlier distributed-model effort encountered exactly these calibration difficulties (Betson et al., 2002). Slug and pumping tests sample only the immediate vicinity of a borehole, and the dense borehole networks needed to interpolate a parameter field across a dune system of several square kilometres are rarely available. Yet many dune sites are monitored, often over decades, by exactly the kind of low-cost instrumentation that distributed models cannot directly exploit: networks of manual dipwells read at monthly intervals, alongside a nearby meteorological station. The methodological question this paper addresses is whether that routine monitoring record, on its own, is sufficient to resolve the hydrogeological architecture of a dune aquifer — its storage, recharge sensitivity and drainage behaviour — with enough spatial detail to support conservation interpretation, and without recourse to coring, hydraulic testing or calibrated continuous-flow modelling.

We develop and demonstrate such a framework. Its core is a three-term state-space model (SSM) fitted independently at each well, in which the monthly change in water-table position is partitioned into a recharge term proportional to rainfall, an atmospheric-draw term proportional to potential evapotranspiration, and a head-dependent drainage term proportional to the water-table displacement above a fixed reference datum. The displacement formulation is the design decision that makes the framework transferable: by referencing drainage to a datum below ground rather than to the ground surface, it captures the head-dependence of lateral discharge without requiring site-specific calibration of drainage geometry, and it resolves a physically meaningful drainage coefficient at every well, including the low-gradient forest interior where a surface-referenced formulation is borderline. The three fitted coefficients are physically interpretable — recharge sensitivity, atmospheric draw and proportional drainage decay — and require only monthly rainfall and temperature from a single station as forcing.

Around this core we assemble a set of complementary, independently derived characterizations that cross-validate one another. Hierarchical clustering of hydrograph *dynamics* — using a correlation-based distance that isolates temporal behaviour from absolute water-table elevation — partitions the network into spatially coherent hydrogeological units before any model is fitted, providing a structure that the mechanistic coefficients must then independently reproduce. A pattern-matching affinity analysis extends the classification to wells with shorter records and identifies the gradational transition zones between units. The SSM is benchmarked against a traditional linear model under iterative simulation, and the spatial pattern of the predictive gain itself becomes a non-invasive diagnostic of where nonlinear storage dynamics govern the aquifer. A water-balance decomposition built directly from the fitted coefficients quantifies the recharge–loss partition at each cluster under identical climate forcing, and an independent water-table-fluctuation estimate of specific yield tests whether the behavioural contrasts between units reflect genuine differences in subsurface storage or boundary conditions imposed at the surface. Finally, interpolation of the per-well coefficients yields continuous maps of recharge sensitivity, atmospheric draw and drainage rate, a mean head surface with its associated flow field, and a water-balance residual field that makes explicit where the three-term model closes and where the three terms are sufficient.

We also resolve an external forcing that conventional climate-trend analysis at a dune site can misattribute. Where a shoreline is actively eroding, the progressive landward migration of the seaward drainage boundary deepens the water table near the coast independently of climate, imposing a spatial gradient on long-term summer-minimum trends. We fit a physics-based, network-scale regression of water-table trend against distance from the eroding shoreline, and use it to partition each cluster's summer-minimum decline into a coastal-retreat component and a climate-driven background — a distinction that matters both for interpreting the aquifer's long-term trajectory and for any subsequent attribution of trends to climate alone.

The framework is demonstrated at Newborough Warren, a coastal dune system on Anglesey, north-west Wales, monitored over twenty-one years by a network of eighty-eight dipwells (eighty-nine monitoring points, including the Llyn Rhos-Ddu lake gauge) spanning the full geomorphological range of the site — from lake-margin and open-dune slacks to a mature coniferous plantation on the elevated landward flank. The density and length of this record make it an unusually well-suited proving ground: it is large enough to test whether the framework recovers structure that is independently visible in the raw hydrographs, and varied enough to span the contrast between a shallow, fast-cycling eastern aquifer and a deep, sluggish western and forested interior.

The objectives of this paper are therefore: (i) to set out the state-space characterization framework and its displacement formulation; (ii) to demonstrate that hydrograph-dynamics clustering, mechanistic SSM coefficients, model-benchmarking diagnostics, water-balance decomposition and specific-yield estimation converge on a single, internally consistent picture of aquifer architecture; (iii) to map that architecture continuously across the site and identify where the three-term water balance closes and where it does not; (iv) to quantify the coastal-retreat forcing on long-term water-table trends; and (v) to assess the framework's transferability to other dune systems monitored by routine dipwell networks. The hydrological consequences of the two management interventions undertaken at the site — dune scraping and a partial plantation clearfell — are quantified separately within a before-after-control-impact design and are reported in a companion paper (Hollingham, 2026b); they are not treated here, where the plantation enters only as a land-cover boundary condition on the aquifer characterization. The intervention dates — the December 2017 clearfell and the dune-scrape years — are the author's recollection, from twenty-one years monitoring the site; the April 2015 CEH36 scrape is separately documented, and no documentary record of the operations has been obtained.

2. Study site

Newborough Warren occupies the south-western tip of the Isle of Anglesey in north-west Wales (Figure 1). The modern dune system is a late-Holocene feature, the present dunes having developed over previously cleared agricultural land following storms in 1330 (Ranwell, 1958; Rhind et al., 2001). The site has no perennial surface drainage; flow lines shown on later figures are DEM-derived flow-accumulation paths included for spatial reference. The dune field is underlain by weakly permeable glacial till, which forms the effective base of the sand aquifer. A bedrock ridge, demarked by the 20 m AOD contour and trending south-south-west to north-north-east, forms the northern boundary of the study area and outcrops within the northern plantation. The hydrological study area, defined by the extent of the DEM-derived flow network, covers approximately 1,170 ha; the dipwell network spans a core zone of roughly 720 ha within it, lying entirely on the south-eastern flank of the ridge. Dune terrain on the north-western flank, draining toward the Malltraeth estuary, is hydrologically separate and is not considered here.

The aquifer south-east of the ridge is divided by its substrate into two contrasting blocks (Stratford et al., 2007). In the east, shallow sand over till and estuarine deposits constrains storage and produces a responsive, flashy water-table regime; in the west, deep aeolian sand accumulation over a broader platform permits substantial groundwater storage and strongly buffered seasonal fluctuations. Whether this east–west contrast reflects two distinct substrate bodies or a single estuarine-clay base that outcrops on the higher eastern and southern ground and dips beneath the surface toward Caernarfon Bay — thinning the sand column in the east and thickening it toward the south-west — is not resolved by the present data (Section 5.1). The aquifer discharges principally across the south-western foreshore of Caernarfon Bay and the southern and eastern coast of the Menai Strait. A small lake within the warren, Llyn Rhos-Ddu, acts as a near-fixed-head body and is the dominant local drainage sink for the immediately adjacent wells, receiving groundwater throughout the year as the regional water table maintains its head above the lake surface (Stratford et al., 2007).

The northern part of the study area is managed plantation dominated by mature Corsican pine. The plantation enters the present analysis solely as a land-cover boundary condition: its canopy intercepts a fraction of incident rainfall before it reaches the ground, and this interception is treated below as a modifier of the surface input to the forested clusters rather than as a management variable. The dune-slack communities of the open system are classified under the National Vegetation Classification as SD15b (wet slack) and SD16 (dry slack); their water-table requirements (Curreli et al., 2013) provide the ecological reference frame for the site but are not used analytically in this paper.

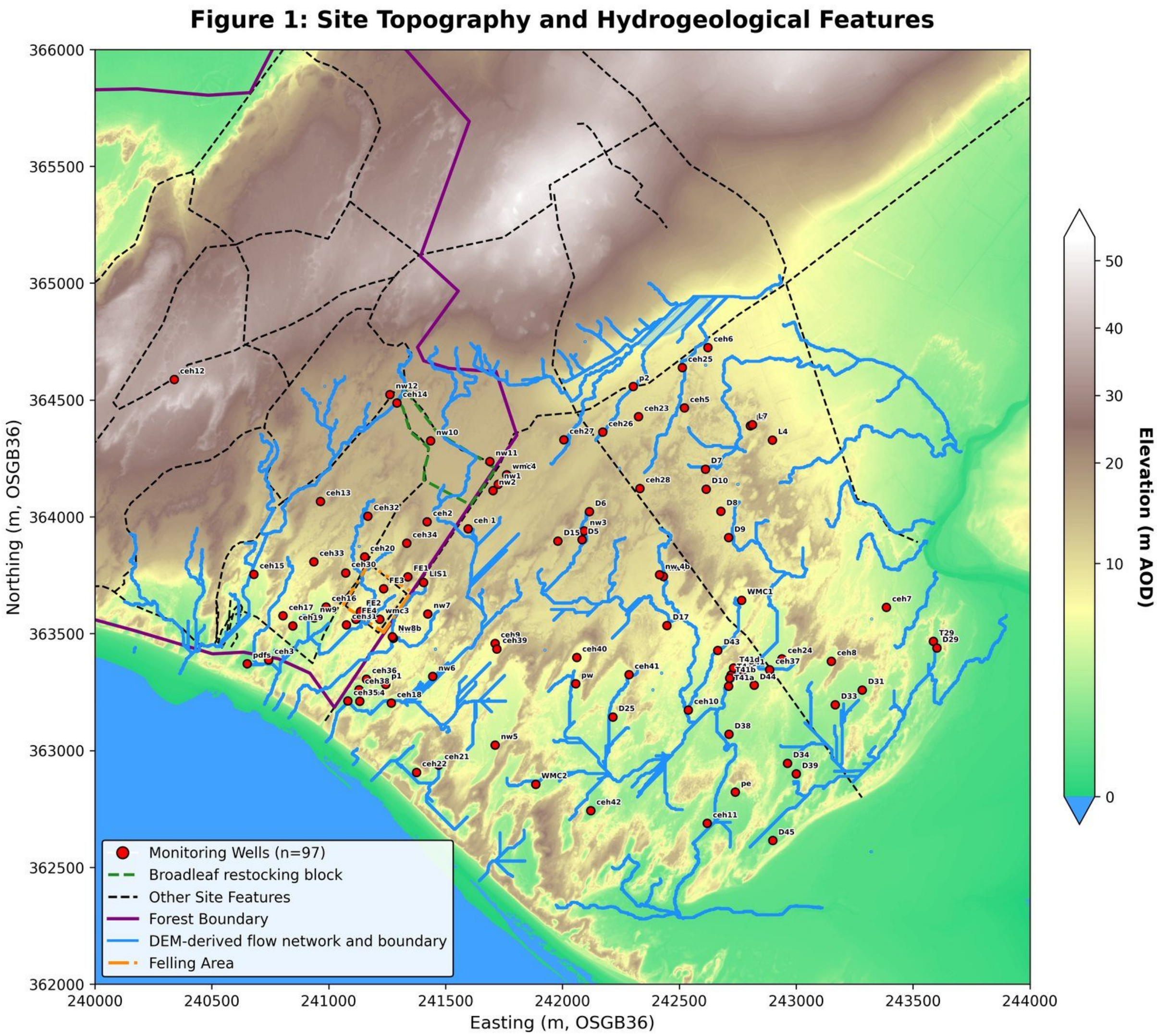

Figure 1. *Site topography, geology and monitoring network at Newborough Warren. (Digital elevation model with rock ridge, forest boundary, watershed boundary, and a flow network derived algorithmically from the DEM for orientation only — no perennial surface streams are present on site. LiDAR data: © NRW & OS.)*

3. Data and methods

3.1 Groundwater and climate records

The primary observational dataset is a network of manually read dipwells, with water-table elevation recorded at approximately monthly intervals. Each reading is taken at the end of the month and represents the water level for the month just ended; readings are bucketed to a monthly timestamp by field convention, with a reading on or before the fifteenth of a month assigned to the preceding month and a later reading assigned to the same month. All monthly series therefore index the month to which the reading belongs rather than the day on which it was taken. Of the full monitoring array, wells with sufficient record length form the reference network used for model fitting; the remainder, with shorter records, form an extended network classified by pattern matching (Section 3.3). Water levels are expressed both in metres above ordnance datum and, for the displacement formulation, as depth below the ground surface at each well.

Climate forcing is taken from the Royal Air Force Valley station. Monthly precipitation totals are used directly. Potential evapotranspiration is computed by the Thornthwaite method from monthly mean temperature and station latitude, with the heat index summed over the trailing twelve months rather than the calendar year so that it is defined for a record that does not end in December (Supplementary Methods S2); this is an empirical index of atmospheric evaporative demand and does not define a specific reference surface in the manner of Penman–Monteith, a property that bears on the interpretation of the atmospheric-draw coefficient (Section 3.4). Rainfall and PET are monthly totals for the same calendar month as the water-table change they are aligned against; no rainfall lag is applied, consistent with the end-of-month measurement convention.

3.2 Hierarchical clustering of hydrograph dynamics

The reference network is partitioned into hydrogeologically distinct zones by agglomerative hierarchical clustering with Ward’s minimum-variance linkage (Ward, 1963). The distance metric is the Pearson correlation distance between standardized well hydrographs, which compares the *shape* of the seasonal and inter-annual water-table dynamics while discarding absolute elevation; two wells that fluctuate in synchrony are treated as similar regardless of their mean water-table depth. This isolates the dynamic hydrogeological signature — the timing and relative amplitude of filling and draining — from the topographic offset between wells.

The number of clusters is selected by a dual-metric approach combining the Ward's merge-distance profile, which identifies the partition at which further agglomeration incurs a disproportionate increase in within-cluster variance, with the mean silhouette coefficient (Rousseeuw, 1987), which favours partitions with well-separated, internally coherent groups. Seasonal amplitude — the long-term mean difference between winter maximum and summer minimum at each cluster centroid — is retained as a secondary descriptor characterizing each unit but is not used in the clustering itself.

3.3 Network extension by Pearson affinity analysis

The primary classification, derived from the reference network, is extended to the wells with shorter records by Pearson affinity analysis. Unlike the mechanistic models, this stage operates on the raw observed time series without model fitting: each well's hydrograph is correlated against every cluster centroid, and the well is assigned to the cluster with which it correlates most strongly. The correlation delta between the assigned and next-strongest cluster classifies membership into tiers, distinguishing core members with an unambiguous dominant affinity from wells with closely competing affinities between neighbouring clusters. A multi-cluster affinity flag identifies wells whose hydrographs are nearly equally well matched by several centroids; the spatial concentration of these flags delineates the gradational transition zones where the boundaries between hydrogeological units are diffuse rather than sharp. The extended-network wells are classified for spatial completeness and are not used in the cluster-level mechanistic fits.

3.4 State-space model and the displacement formulation

The mechanistic core of the framework is a three-term linear state-space model relating the monthly change in water-table position to climate forcing and to the antecedent head. For a given well or cluster centroid, the model is

$$\Delta h(t) = \beta_1 \cdot P(t) - \beta_2 \cdot PET(t) - \beta_3 \cdot h\_disp(t-1)$$

where $\Delta h(t)$ is the change in water-table position during month t, $P(t)$ and $PET(t)$ are the month-t precipitation and potential evapotranspiration totals, and $h\_disp(t-1)$ is the water-table displacement above the drainage datum at the end of the preceding month. The three coefficients are constrained to enter with physically correct sign and are reported as positive quantities: β_1 (recharge sensitivity) is the water-table rise per unit rainfall; β_2 (atmospheric draw) is the decline per unit PET; and β_3 (proportional drainage decay) is the fraction of antecedent displacement dissipated by lateral drainage each month.

The drainage term is referenced to a *displacement* above a fixed datum rather than to the absolute or ground-surface-referenced head:

$$h\_disp = z_0 + h\_depth$$

where z_0 is the drainage datum (a uniform depth below the ground surface, set at 3.7 m for the primary analysis) and h_depth is the water-table position in the negative-below-ground convention. Two features of this specification are central. First, the drainage term uses the displacement at the *end of the previous month* — the start of the current month — rather than the current month's level. Using the contemporaneous level would introduce simultaneity bias, since that level is itself the result of the drainage being modelled; physically, the discharge during a month is driven by the head at the start of the month. Second, referencing drainage to a sub-surface datum rather than to the ground surface allows the head-dependent drainage response to be resolved consistently across wells of very different mean water-table depth, including the low-gradient forest interior where a surface-referenced drainage term is weak and borderline. The sensitivity of the fit to the datum choice is examined by independently optimizing z_0 at each well over a sweep of candidate depths and comparing the resulting fit against the uniform datum (Section 4.7).

The model is fitted by ordinary least squares without an intercept at each reference well and at each cluster centroid hydrograph. The recharge sensitivity of each unit is summarized by the lumped catchment storage coefficient, LCSC = $100/\beta_1$, expressing the depth of rainfall required to raise the water table by a fixed increment; a lower LCSC denotes greater recharge sensitivity. Because the fitted coefficients fold together vegetation effects, specific yield and soil-moisture dynamics, and because no intercept is fitted, the recharge term in an open-boundary system of this kind can absorb unmeasured boundary fluxes into β_1 (Young, 2011; Knotters and van Walsum, 1997); this is addressed explicitly through the residual-field analysis (Section 3.10) and the specific-yield estimation (Section 3.8).

A depth-dependent extension, in which β_2 is replaced by $\beta_2 \cdot \exp(-\kappa d)$ to allow atmospheric draw to decline as the capillary fringe retreats from the root zone with increasing water-table depth d, was evaluated at cluster level by grid search over κ. Because the extension introduces a nonlinearity that would propagate into the closed-form analytical outputs derived elsewhere, it is retained as a diagnostic of depth-coupling rather than adopted as the operational model; the fixed-β_2 SSM is used throughout, understood as an effective depth-averaged parameterization.

The model is fitted at two scales for two purposes. The mechanistic characterization is fitted to each cluster-mean hydrograph, where averaging the member records suppresses single-well noise and local topographic idiosyncrasy and returns one coefficient set per hydrogeological zone — appropriate because the zones are defined so that their members share a dynamic signature, and because cross-zone comparison of mechanism is the aim. The continuous coefficient, head and residual surfaces are instead built from the model fitted independently at each reference well, since their purpose is the within-zone spatial structure that a cluster mean would remove by construction. The trade-off is explicit: cluster fitting gains robustness at the cost of within-zone variability, which is substantial — recharge sensitivity varies about two-fold within the Western Residual alone (Table 6) — so a cluster coefficient is a centroid, not a property of every member; and it depends on the partition, since a misassigned or genuinely gradational well biases the mean it enters, a dependence the per-well fits avoid. The two scales are used as complementary throughout, cluster coefficients for mechanism and per-well coefficients for spatial distribution, with the per-well ranges (Table 6) reported alongside the cluster means rather than subsumed into them.

3.5 Model benchmarking against a traditional linear model

The predictive value of the explicit storage-decay term is assessed by benchmarking the SSM against a traditional linear model (TLM) that omits the head-dependent drainage term, fitting the water-table change as a function of climate forcing alone. Both models are fitted at every reference well and compared in two modes. In one-step diagnostic mode, each prediction is corrected by the observed level from the previous month, and the two models are compared by coefficient of determination. In iterative simulation mode, the model is run forward from an initial condition without observational correction, propagating its own predictions, and the two models are compared by Nash–Sutcliffe efficiency. The iterative comparison is the discriminating test, because it exposes the cumulative drift that arises when storage memory is omitted. The spatial pattern of the iterative NSE gain is mapped across the network as a diagnostic of where head-dependent drainage memory is indispensable to prediction and where the aquifer approximates a linear climate response.

3.6 Spatial coefficient mapping and within-forest regression

The per-well SSM coefficients are interpolated across the site by inverse-distance weighting over a regular grid built on a LiDAR-derived digital elevation model, with ridge cells and non-aquifer areas masked. This yields continuous surfaces of recharge sensitivity, atmospheric draw and drainage rate, together with a model-fit-quality surface, providing a spatially resolved expression of aquifer behaviour that complements the cluster-level summaries.

Within the forested zone, where the canopy is broadly uniform but the per-well coefficients span much of the site-wide range, the per-well coefficients are regressed against elevation, distance from the ridge and easting to test whether the within-forest variation tracks topographic position and water-table depth or canopy differences. Because the subsurface beneath the plantation is not cored, elevation and position serve as proxies for substrate, and the regression isolates this topographic control on the mechanistic coefficients beneath a constant land cover.

3.7 Mean monthly water-balance decomposition

The fitted cluster coefficients are used to decompose the mean monthly water balance of each cluster into its component fluxes under identical climate forcing. Recharge is computed as β_1 times mean monthly precipitation, atmospheric draw as β_2 times mean monthly PET, and gravity drainage as β_3 times the mean displacement above the datum. All terms are expressed in metres of water-table change per month — the natural unit of the state-space model — and are directly comparable across clusters that receive the same climate input. Canopy interception is not added as a separate loss term: in the forested clusters it is already embedded implicitly in the lower fitted β_1 and β_2, and adding it explicitly would double-count losses. The net monthly balance — recharge minus total losses — quantifies the residual that would be required from an unmeasured boundary input to maintain long-run equilibrium.

An indicative volumetric conversion is presented alongside the head-space decomposition using assumed specific-yield values appropriate to the substrate of each cluster, with the empirical specific-yield estimates of Section 3.8 used to bracket the volumetric fluxes from above. For the forested clusters, canopy interception is applied as a reduction of the surface input following the site-specific interception fraction (Freeman, 2008) — a single-plot, single-season measurement, and the only site-specific value available — reducing the boundary input rather than being additive to losses. The evapotranspiration–drainage partition is bracketed by two independent methods — the SSM headspace ratio and a seasonal recession-curve analysis — and reported as a mid-point with an associated uncertainty range.

3.8 Specific-yield estimation by the water-table-fluctuation method

An independent estimate of specific yield is derived from the dipwell record by the water-table-fluctuation method (Healy and Cook, 2002), in which specific yield is the ratio of net recharge to the corresponding water-table rise. The method is applied to the cluster-mean hydrographs, retaining only months in which both the water table rose and net recharge was positive above defined thresholds, and constraining event-level estimates to a physically plausible range. Because monthly data cannot isolate individual storm events and the method at monthly resolution is known to conflate gravity drainage with capillary-fringe release, the resulting estimates are interpreted as upper bounds on effective storage rather than as point estimates (Healy and Cook, 2002; Scanlon et al., 2002).

For the forested clusters a canopy-interception correction is applied to the precipitation term before computing net recharge, reducing only the rainfall input and leaving the Thornthwaite PET term unchanged. PET is an energy-based atmospheric-demand index independent of land cover; reducing it to account for canopy-evaporated interception would conflate atmospheric demand with realized evapotranspiration and remove the same quantity from both sides of the balance. The corrected forest estimates are compared against the open-dune range to test whether the attenuated water-table response beneath the canopy reflects the surface boundary condition or a genuinely different subsurface storage. The per-well event-median specific yields are interpolated across the site to assess whether the spatial distribution of storage is uniform or shows discontinuities coinciding with the cluster margins.

3.9 Network-scale coastal-retreat gradient regression

Where a shoreline is actively eroding, the landward migration of the seaward drainage boundary acts as a slowly advancing line sink: it lowers the ground surface relative to the water table, raising the displacement head at adjacent wells and steepening the seaward hydraulic gradient, so that the near-coast water table deepens over time independently of climate along a Dupuit drawdown curve (Fetter, 2001). This mechanism — an eroding dune margin drawing down its own hinterland — is documented at the retreating Whiteford Burrows dune system in South Wales (Robins, Pye and Wallace, 2013), and imposes a spatial gradient on long-term summer-minimum trends. To partition each cluster’s observed summer-minimum decline into a coastal-retreat component and a climate-driven background, a network-scale, physics-based non-linear regression is fitted to per-well water-table trends against distance from the eroding shoreline. The perpendicular distance from each well to the eroding Caernarfon Bay high-water mark is computed within the pipeline as the minimum distance to a committed coastline polyline (OpenStreetMap mean-high-water, EPSG:27700) restricted to the west-facing Newborough shoreline, with the Menai Strait excluded as a non-eroding boundary.

Two functional forms are fitted: a Dupuit–Forchheimer steady-state strip-aquifer form, in which the coastal anomaly declines linearly to a far-field background and is capped at zero; and an exponential-decay form representing a diffusive transient response. These are the steady-state and transient limits of the same drawdown physics — a Dupuit drawdown curve from the migrating boundary, its inland reach L_cg set by the aquifer's hydraulic diffusivity D = Kb/Sy, the same length-scale physics that governs the interior head-loss reach of the forest block. Both are fitted by profile non-linear least squares with well and month fixed effects and a cumulative-water-balance covariate absorbed by within-well demeaning, with model selection between the two forms by the Akaike information criterion (Akaike, 1974). Three nested specifications are fitted to test robustness to forest-cover inclusion: the full network excluding the clearfell-zone wells; a forest-free network dropping all forested wells (C4 Main Forest and C5 Coastal Forest); and a single-cluster specification using only the western non-forested cluster (C3 Western Residual), with the far-field background held to the network value because that cluster contains no well at the coast and an unconstrained fit on its restricted distance range would be under-identified. The forest-free linear-with-cutoff fit is taken as the headline. A fourth fit controls forest cover explicitly rather than by exclusion, refitting the full network with a canopy × time term; the coast-edge amplitude is essentially unchanged and the absorbed canopy effect is a separate -9.0 ± 2.5 mm yr^{-1} deepening under pine, confirming forest cover does not drive the gradient (Supplementary Methods S13.2). The headline fit is then applied to decompose each cluster's (C1–C5) summer-minimum slope into gradient and climate components, quantifying the contribution of coastal retreat to long-term water-table trends across the network.

3.10 Spatial interpolation, flow field and water-balance residual field

A mean water-table surface is constructed by inverse-distance interpolation of per-well mean elevations to a regular grid on the British National Grid, masked to the sea-boundary extent. No elevation mask is applied, because the water table is a continuous surface independent of the overlying topography. Normalized Darcy flow-direction vectors are computed from the head-gradient field using a single representative hydraulic conductivity (Betson et al., 2002) and an interpolated aquifer-thickness surface; these are reported as directional context for the interpolated head field rather than as the output of a calibrated continuous-flow model (Bear, 1972; Freeze and Cherry, 1979).

The water-balance residual field quantifies, at each reference well, the degree to which the three-term SSM closes the long-term water balance. The residual is computed from the fitted coefficients as the difference between modelled mean losses (atmospheric draw plus drainage) and modelled mean recharge, with the gross rainfall at all wells, including the forested ones: interception is already carried in $\beta_2 \cdot \overline{PET}$, since the SSM is fitted on gross rainfall and above-canopy Thornthwaite PET, and subtracting it from rainfall as well would double-count it. Positive residuals indicate locations where modelled losses exceed modelled recharge — an inadequacy in the three-term model at that location, or in principle an unmodelled input; near-zero residuals indicate that the three-term balance closes adequately. The field is interpolated to a grid with sea-boundary anchors and ridge cells masked. It is interpreted as a structural diagnostic of where the model's three terms are sufficient, not as a quantified map of lateral inflow, since the residual also absorbs nonlinear soil-moisture storage dynamics, coefficient sensitivity and any systematic bias in the Thornthwaite PET estimate. The mechanistic attribution of the residual pattern is tested by an independent cross-correlation lag analysis for distance-dependent transport from the ridge and a seasonal climatology analysis for unmodelled summer evaporative demand.

3.11 Computational implementation

All data pre-processing, statistical modelling and spatial mapping were implemented as a sequential, numbered Python pipeline, with time-series management in pandas and NumPy, model fitting in statsmodels, and spatial interpolation and rendering over a LiDAR DEM. The state-space alignment and displacement calculation are implemented once in a shared model-utilities module and called by every script that fits the SSM, ensuring that all downstream analyses are consistent with the cluster-level fits. The pipeline and its outputs are reproducible by running the published scripts in sequence.

4. Results

4.1 Climate and well-network characterization

The RAF Valley record establishes that Newborough Warren operates under a persistent annual water surplus, with mean annual precipitation of approximately 891 mm against mean annual Thornthwaite PET of approximately 652 mm over the monitoring period, a study-period P/PET ratio of about 1.36 (Figure 2). The surplus is strongly seasonal in its expression: winter rainfall exceeds evaporative demand by a wide margin, while summer PET exceeds rainfall, concentrating atmospheric demand on the aquifer during the period of lowest recharge. Inter-annual variability in the water balance is driven almost entirely by precipitation rather than by evaporative demand, which shows strong seasonality but negligible inter-annual trend. The post-2013 period has run warmer than the earlier summer-temperature baseline (Figure 3), intensifying the summer deficit.

The reference network comprises 66 wells with monthly water-table records over the 2005–2026 monitoring period, supplemented by 22 extended-network wells with shorter records classified by pattern matching (Section 4.3). Records run to a median length well above a decade, giving the spatial density needed to distinguish cluster-scale behaviour from site-wide climate forcing.

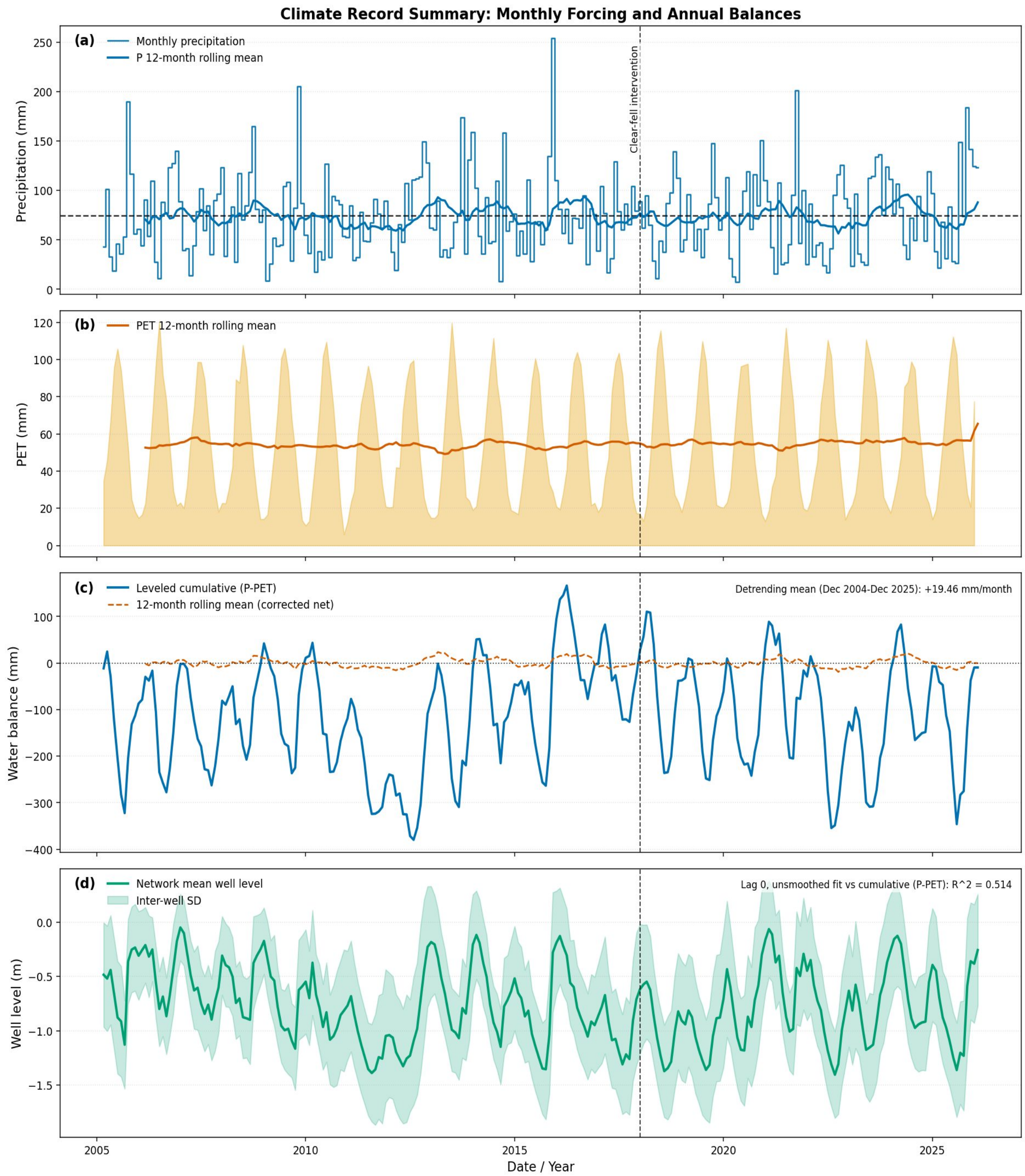

Figure 2. *Climate forcing over the monitoring period (2005–2026) from RAF Valley: monthly precipitation and Thornthwaite PET, with cumulative water surplus. The full ~95-year station record is not shown here; the long-term warming context is given in Figure 3.*

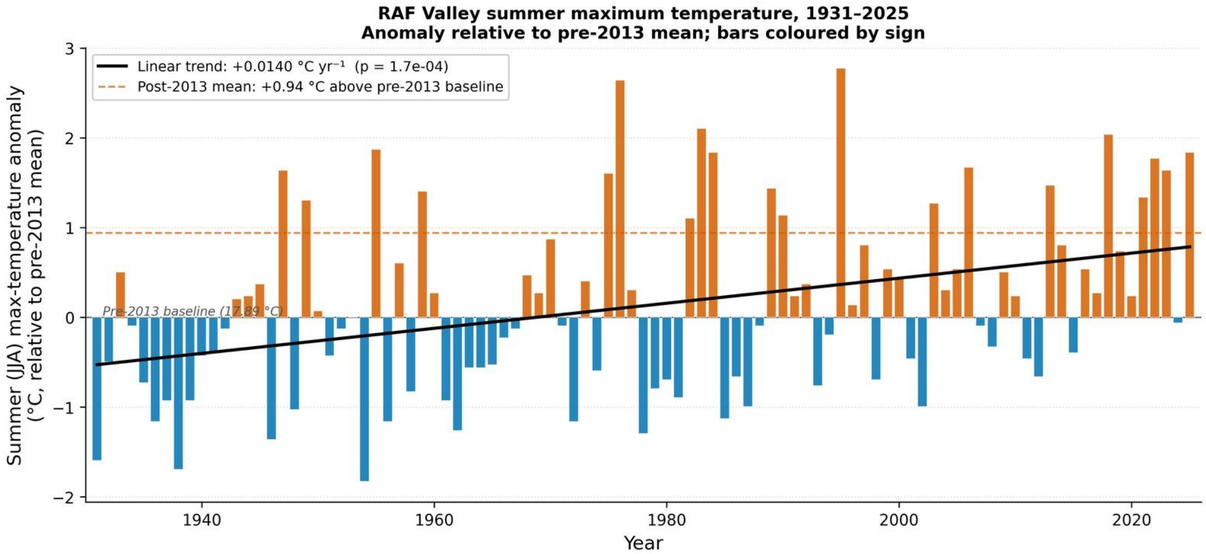

Figure 3. *Summer-temperature record showing the post-2013 warming relative to the earlier baseline.*

4.2 Network clustering and spatial architecture

Hierarchical clustering of the standardized hydrograph dynamics, using the Pearson correlation distance and Ward's linkage, resolves five spatially coherent units under the dual-metric selection of k (Figure 4). The partition separates the network into a lake-margin unit (C1, Lake Edge), an open-dune unit (C2, Dune), a western unit on deeper sand (C3, Western Residual), and two units beneath the Corsican pine plantation distinguished by topographic position (C4, Main Forest, on the elevated ridge flank; C5, Coastal Forest, on the lower coastal fringe) (Figure 5). The cluster sizes are C1 = 7, C2 = 24, C3 = 21, C4 = 9 and C5 = 5 wells. The five units recover the broad substrate contrast between the responsive, storage-limited east and the buffered, deeper-sand south-west (Stratford et al., 2007), the latter (C3) gradational toward both the open dune and the forest rather than a sharply bounded block, with the two forested units segregating from this south-western dune despite their inferred substrate similarity. The cluster-centroid hydrographs and seasonal amplitudes are shown in Figure 6.

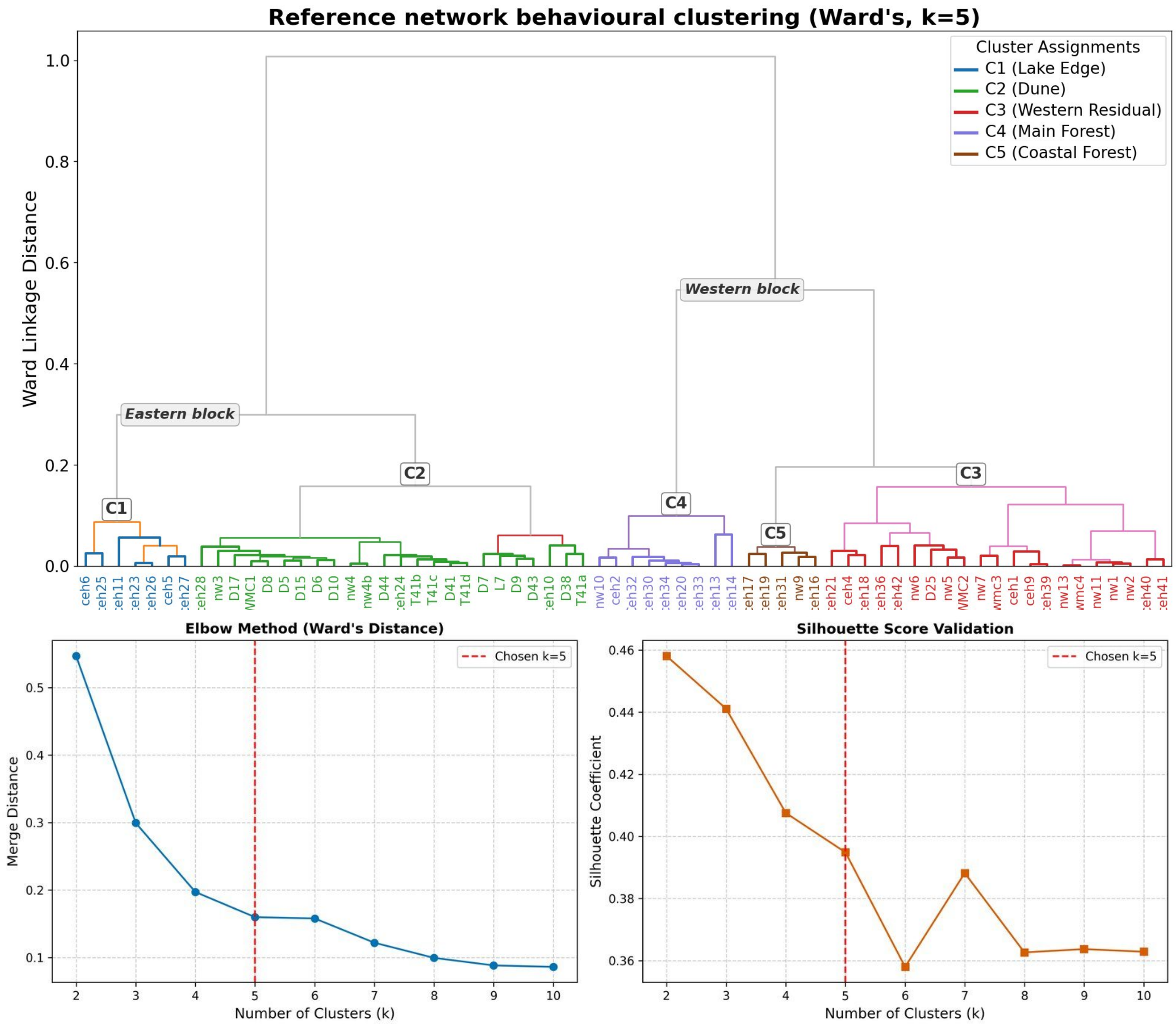

Figure 4. *Hierarchical clustering of hydrograph dynamics: Ward's dendrogram (upper) and the dual-metric cluster-number selection (lower).*

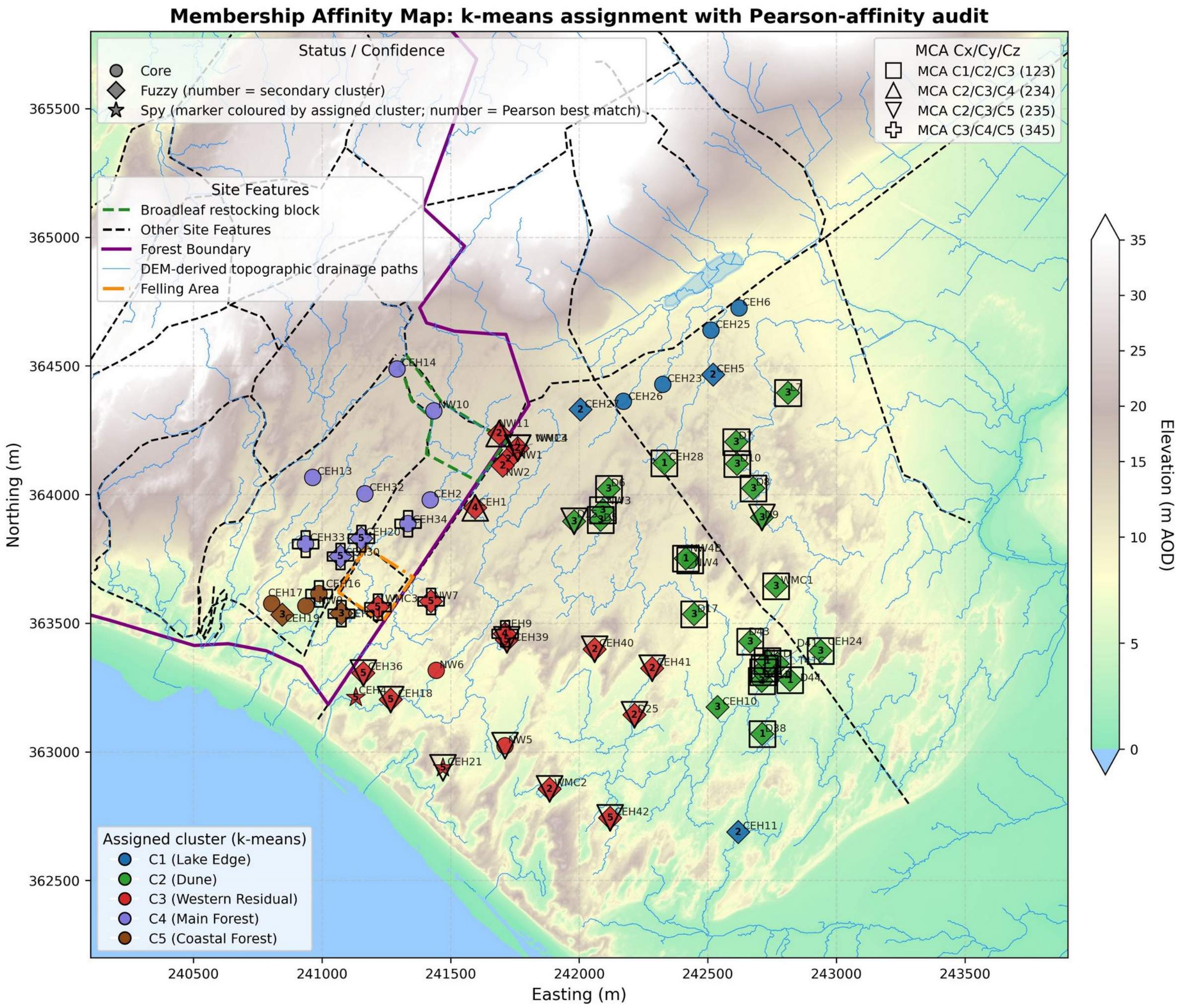

Figure 5. *Spatial distribution of the five hydrogeological clusters (Pearson affinity validation), reference network (66 wells). Points coloured by k-means assignment; circles = Core (Δr > 0.05), diamonds = Fuzzy (Δr < 0.05; number inside diamond = secondary cluster), stars = Spy (Pearson best-match ≠ assigned; number inside star = best-match cluster). Hollow overlays = Multi-Cluster Affinity (r > 0.90 with ≥3 clusters). LiDAR: © NRW & OS.*

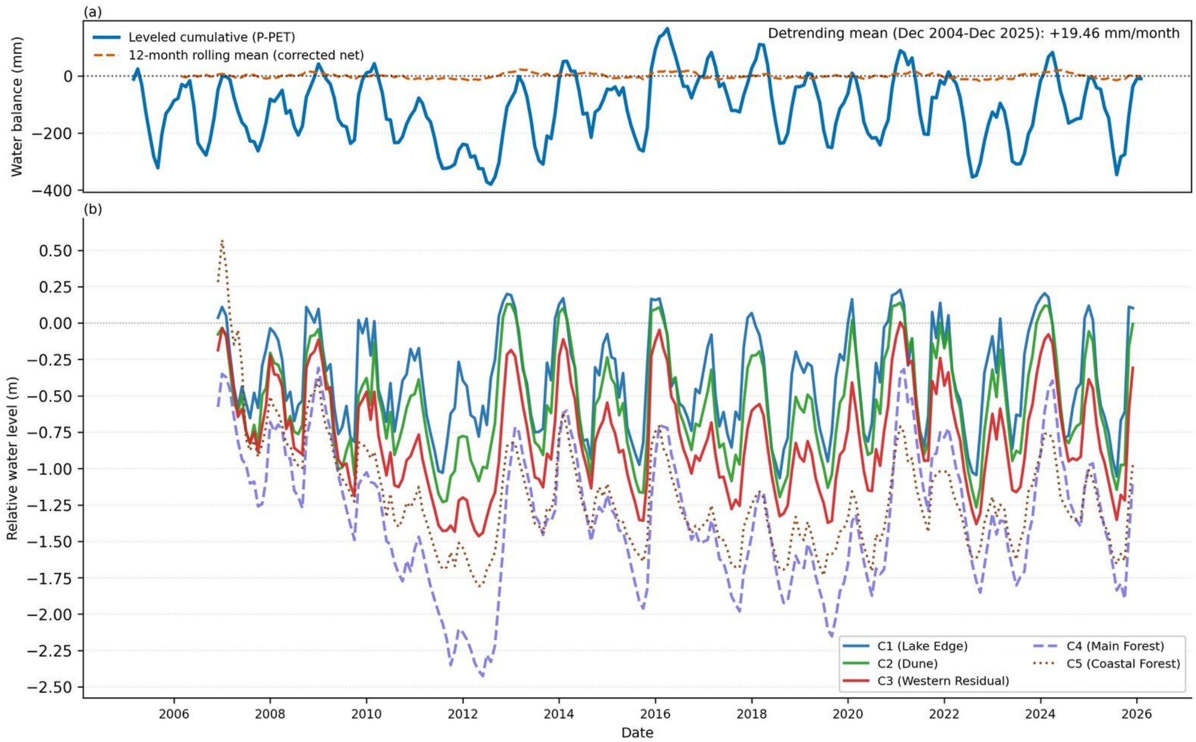

Figure 6. *Cluster-centroid hydrographs and seasonal amplitudes, 2005–2026.*

4.3 Affinity validation of the cluster structure

The Pearson affinity analysis confirms the primary membership of the great majority of reference wells. Membership separates into tiers by the correlation delta between the assigned and next-strongest cluster: a minority of wells are core members with an unambiguous dominant affinity, while many show closely competing affinities between neighbouring units, consistent with gradational rather than sharp boundaries in a contiguous, highly transmissive sand aquifer (Rao and Srinivas, 2006). The spatial concentration of multi-cluster-affinity flags along the C1/C2/C3 transition and the forest–dune margin identifies the zones where geological and vegetative controls overlap.

4.4 Cluster mechanistic coefficients

The five clusters are characterized mechanistically by the displacement-formulation SSM fitted to each cluster-centroid hydrograph (Table 1). All coefficients are statistically significant ($p < 0.003$) at every cluster.

__Table 1.__ Cluster mechanistic characterization. β_1 = recharge sensitivity (mm rise per mm rainfall); β_2 = atmospheric draw (mm decline per mm PET); $-\beta_3$ = proportional drainage decay (fraction of displacement above the 3.7 m datum dissipated per month); LCSC = 100/β_1, the reciprocal of β_1 (mm of rainfall per 100 mm of water-table rise); R^2 = one-step diagnostic fit.

Cluster	n wells	β_1	β_2	$-\beta_3$	LCSC (%)	R^2
C1 Lake Edge	7	4.58	0.92	0.09	21.8	0.73
C2 Dune	24	3.97	1.74	0.06	25.2	0.75
C3 Western Residual	21	3.57	1.81	0.06	28.0	0.81
C4 Main Forest	9	2.48	2.56	0.02	40.4	0.72
C5 Coastal Forest	5	2.43	1.27	0.04	41.2	0.68

Recharge sensitivity β_1 is highest at the Lake Edge (C1: 4.58), indicating that each millimetre of rainfall drives a proportionally larger water-table rise in the shallow-till eastern aquifer than in the Dune (C2: 3.97) or Western Residual (C3: 3.57). The two forested clusters return near-identical and substantially lower recharge sensitivities (C4: 2.48, C5: 2.43), consistent with canopy interception suppressing the rainfall signal reaching the water table. Expressed as LCSC (100/β_1), about 22–25 mm of rainfall raises the water table by 100 mm at the Lake Edge and Dune, 28 mm in the Western Residual, and 40–41 mm beneath the plantation — nearly twice as much rain for the same rise beneath the canopy.

Atmospheric draw β_2 is highest at C4 (2.56) — where it slightly exceeds C4's own β_1 (2.56 against 2.48), so the Main Forest cluster loses marginally more water table per millimetre of PET as it gains per millimetre of rainfall — followed by C3 (1.81) and C2 (1.74), with C5 (1.27) and C1 (0.92) lowest. That C5 carries the same pine canopy as C4 yet returns less than half its atmospheric draw points to topographic position and water-table depth rather than canopy as the dominant control of the cluster-level β_2 contrast (Section 4.8). C1's low β_2 reflects its shallow water table and rapid lateral drainage to the lake, where head-dependent drainage captures the dominant loss.

The drainage decay $-\beta_3$ shows the strongest contrast of the three coefficients: highest at C1 (0.089), where rapid lateral exchange with Llyn Rhos-Ddu drives fast recession, similar at C2 (0.064) and C3 (0.057), lower at C5 (0.045), and lowest at C4 (0.018), where a suppressed recharge pulse and weak lateral discharge produce the most sluggish drainage in the network.

4.5 Water balance and specific yield

The head-space water-balance decomposition (Table 2; Figure 7) closes to within 1.8% of total losses at every cluster, confirming that the three-term displacement SSM captures the principal fluxes without requiring an unmeasured boundary term at the cluster mean. Recharge declines systematically from C1 (0.340 m/month) to the forested clusters (≈0.18 m/month), and total losses track recharge closely at every cluster. The drainage partition varies markedly: drainage accounts for 85% of total losses at C1 but only 24% at C4, where atmospheric draw dominates.

Table 2. *Mean monthly head-space water balance (m/month), 2005–2026. Identical climate forcing at every cluster ($\bar{P} \approx 74.4$ mm/month, $\bar{PET} \approx 54.7$ mm/month).*

Cluster	Recharge	Atm. draw	Drainage	Total loss	Residual	Drain % / ET %
C1 Lake Edge	0.34	0.05	0.29	0.34	−0.004	85 / 15
C2 Dune	0.30	0.09	0.20	0.30	−0.003	68 / 32
C3 Western Residual	0.27	0.10	0.17	0.27	<0.001	63 / 37
C4 Main Forest	0.18	0.14	0.04	0.18	<0.001	24 / 76
C5 Coastal Forest	0.18	0.07	0.11	0.18	−0.003	62 / 38

The indicative volumetric conversion (Table 3) places all clusters under ≈892 mm/yr rainfall, with canopy interception removing ≈214 mm/yr at the forested clusters to leave ≈678 mm/yr of throughfall. The ET/drainage partition mid-point ranges from drainage-dominated at C1 (≈649 mm/yr drainage vs ≈243 mm/yr ET) to ET-dominated at C4 (≈248 vs ≈430 mm/yr).

Table 3. *Indicative annual volumetric water balance (mm/yr). I = canopy interception (24% of P; Freeman, 2008), forested clusters only. ET/drainage are mid-points of two independent partitioning methods.*

Cluster	P	I	P_net	ET (mid)	Drainage (mid)
C1 Lake Edge	892	0	892	243	649
C2 Dune	892	0	892	388	505
C3 Western Residual	892	0	892	334	559
C4 Main Forest	892	214	678	430	248
C5 Coastal Forest	892	214	678	218	461

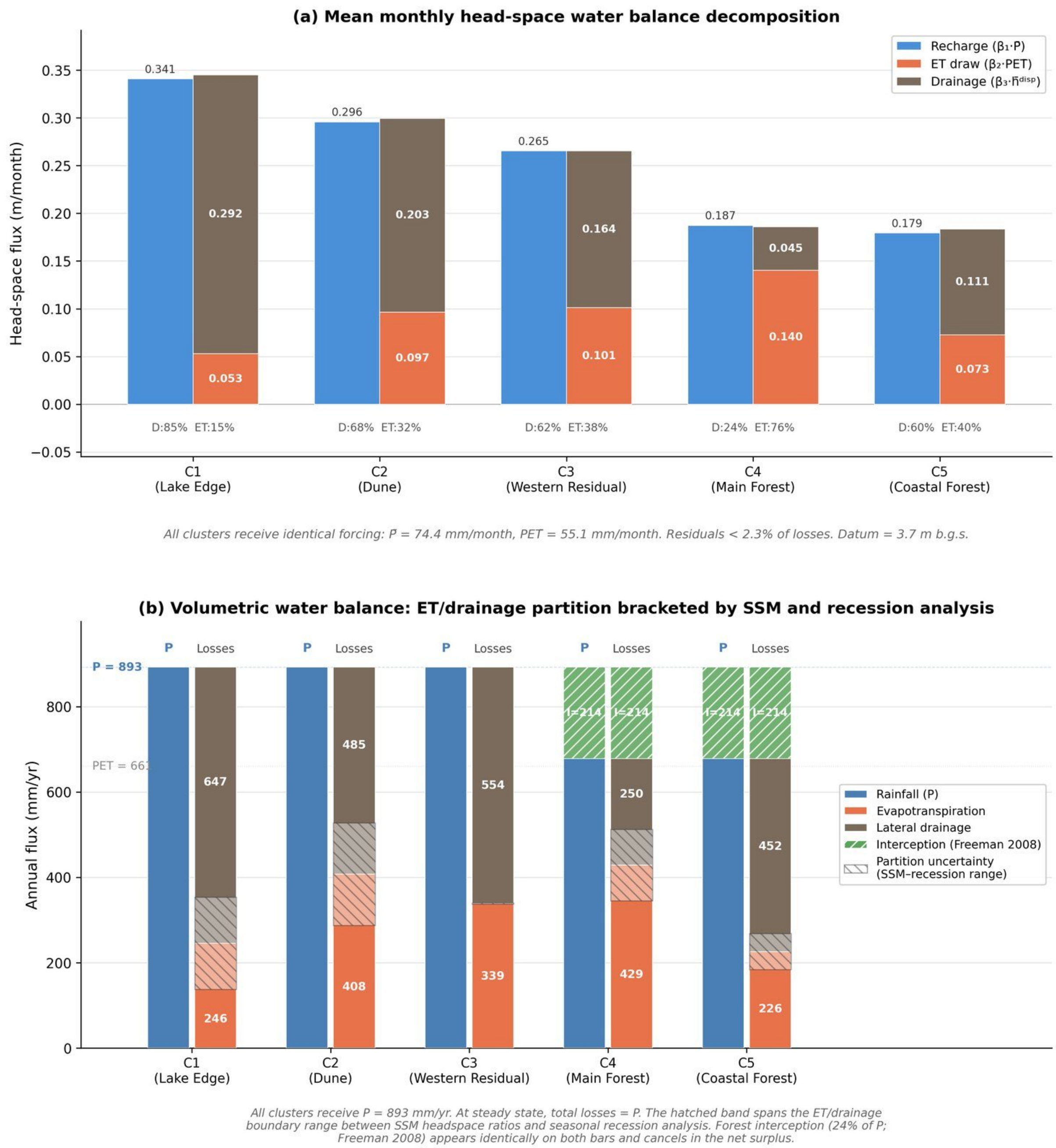

Figure 7. *Water-balance decomposition by cluster: head-space (recharge versus atmospheric draw and drainage) and indicative volumetric partition. In (b), the hatched partition-uncertainty band spans the two independent estimates of the ET/drainage split — the SSM head-space ratio and the seasonal winter/summer recession ratio; it is negligible for C3, where the two methods coincide (both ≈0.60 drainage / 0.40 ET).*

The water-table-fluctuation specific-yield estimates (Table 4; Figure 8) provide an independent test of whether the cluster contrasts reflect storage architecture or surface boundary conditions. Uncorrected forested-cluster medians (C4: 0.312, C5: 0.355) exceed the open-dune range, because gross rainfall overstates the recharge flux beneath the canopy. Applying the canopy-interception correction to the recharge term brings the cluster-level event medians down to 0.260 (C4) and 0.321 (C5): C4 falls cleanly into the open-dune range (0.210–0.327), while C5 sits at its upper edge. Aggregating instead as the median of the per-well estimates gives 0.252 and 0.306; the two aggregations are reported separately throughout and are not interchangeable. Across the three open-dune clusters the per-well specific yields are well described by a planar spatial trend ($n = 52$, $R^2 = 0.613$, residual standard deviation 0.031), rising by 0.071 per kilometre along an azimuth of 234°, so that storage increases steadily toward the south-west. Both components are significant (easting −0.058 per kilometre, northing −0.042 per kilometre, each $p < 0.001$). The trend is fitted to, and reported for, the open-dune network only: the forested wells lie outside its spatial domain and the surface is not extrapolated beneath the plantation. The C4 result indicates that the attenuated water-table response beneath the plantation reflects the surface canopy boundary condition rather than a fundamentally different substrate. The C5 estimate is the least reliable in the network — a majority of its rising-limb events imply an event-level Sy at or above the porosity of dune sand, so its median is constrained by the physical-plausibility limit rather than freely estimated — and it is reported as a weak corroborator only. All WTF medians exceed the assumed values used in the volumetric balance by a factor of two to three and are interpreted as upper bounds (Healy and Cook, 2002; Scanlon et al., 2002).

Table 4. *Specific yield by cluster, water-table-fluctuation method. Event-median Sy; forested clusters reported uncorrected and interception-corrected.*

Cluster	Sy assumed	Sy WTF median (uncorr.)	Sy WTF median (corrected)
C1 Lake Edge	0.08	0.21	—
C2 Dune	0.12	0.27	—
C3 Western Residual	0.12	0.33	—
C4 Main Forest	0.12	0.31	0.26
C5 Coastal Forest	0.12	0.35	0.32

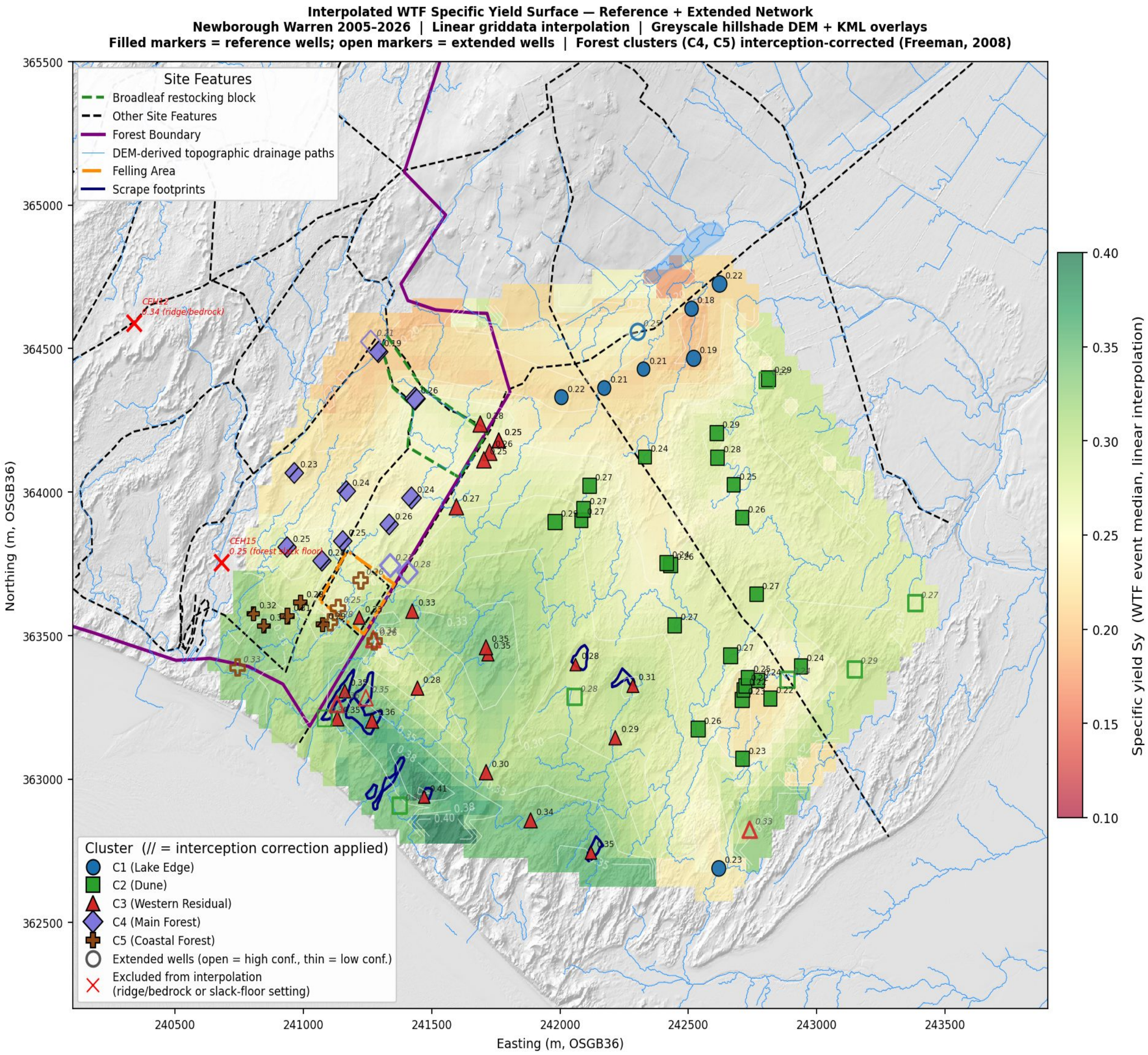

Figure 8. *Interpolated water-table-fluctuation specific-yield surface; forest wells interception-corrected. The distribution varies smoothly, with no discontinuity at the cluster margins. The surface is a spatial diagnostic of a lumped-parameter model, not the output of a calibrated continuous-flow simulation. The interpolation extends over the rock-ridge bedrock outcrop on the northern boundary, which carries no monitoring wells; values shown there are extrapolations from the surrounding network and are not interpretable as model output on bedrock.*

4.6 Model benchmarking: SSM versus traditional linear model

In one-step diagnostic mode the SSM and TLM are nearly indistinguishable (median R^2 0.92 and 0.91), because each prediction is corrected by the previous observed level. The distinction emerges under iterative simulation, where the SSM achieves a median NSE of 0.72 while the TLM returns a median of −0.03. A negative Nash–Sutcliffe efficiency means the model leaves more error than the observed long-term mean would, so at the median well the TLM performs worse than simply predicting the mean water table; the SSM improves on it by +0.82 NSE units and returns a positive iterative NSE at 65 of 66 wells against 30 of 66 for the TLM (Table 5; Figure 9). The explicit β_3 term captures storage-decay memory and prevents the cumulative drift that causes the TLM to diverge. The largest gains occur in the lake-proximal eastern aquifer (e.g. CEH27: TLM −2.18 → SSM 0.68; CEH6: −1.89 → 0.67), where head-dependent drainage is strongest; the smallest occur in the forest interior, where weak drainage makes the storage term nearly dispensable. The spatial pattern of the gain therefore serves as a non-invasive diagnostic of where nonlinear storage dynamics govern the aquifer.

Table 5. *SSM (B) versus traditional linear model (A) benchmarking.*

Metric	TLM (A)	SSM (B)	Δ (B−A)
Median one-step R^2	0.91	0.92	0.00
Median iterative R^2	0.64	0.77	0.14
Median iterative NSE	−0.03	0.72	0.82
Wells with iterative NSE > 0	30 / 66	65 / 66	—

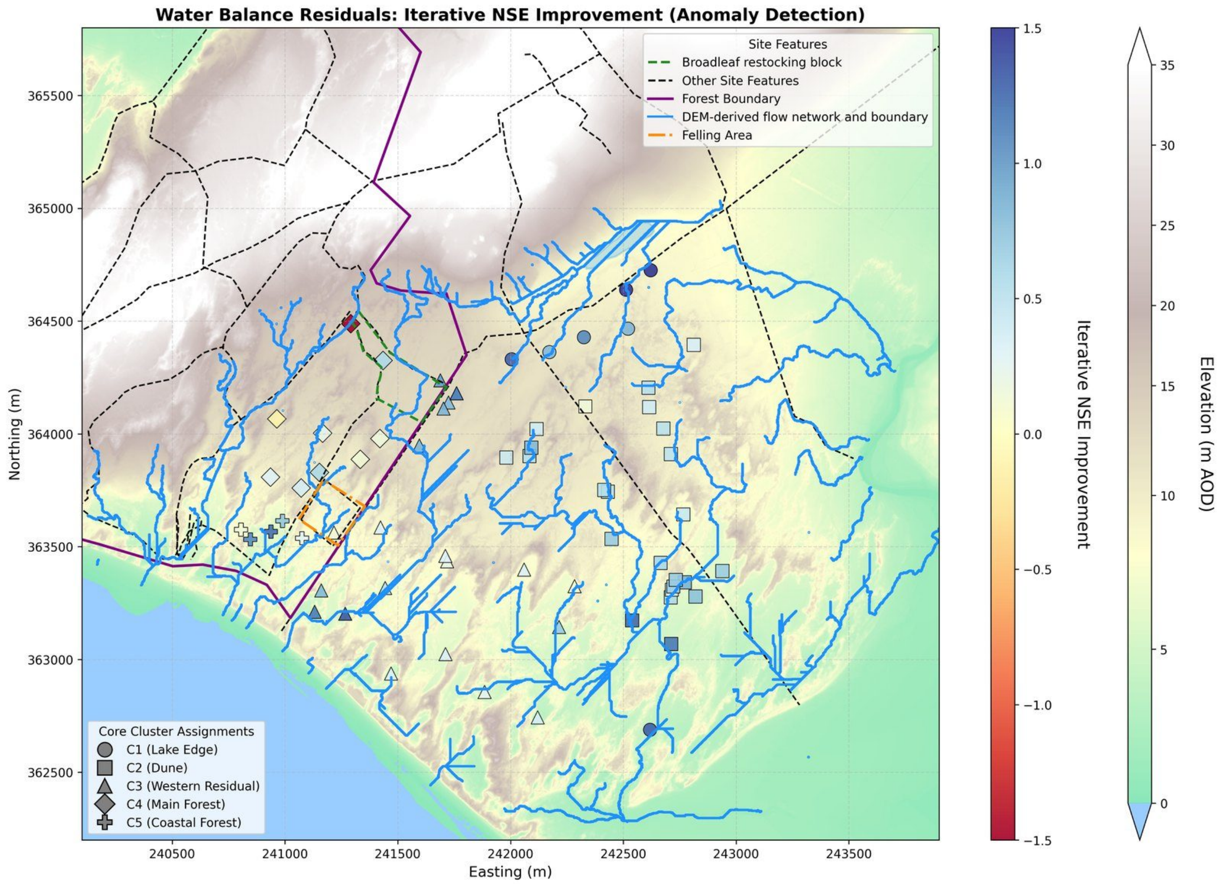

Figure 9. *Spatial pattern of the SSM-over-TLM iterative Nash–Sutcliffe improvement across the reference network.*

4.7 Spatial coefficient structure and drainage half-life

Per-well drainage-datum optimization shows the R^2-maximizing datum varying systematically across the site, from shallow at the Lake Edge to deep in the western and forest interior, but the R^2 gain over the uniform 3.7 m value is negligible across most of the network (Figure 10), confirming the uniform-datum assumption. The shallow-to-deep gradient broadly mirrors the eastward-shallowing substrate, but the near-flat R^2–datum response, and the confounding of the optimal datum with mean water-table depth (below which it must lie), leave it too weakly constrained to be read as recovering the impermeable base. The confounding is direct: across the network the R^2-maximizing datum correlates more strongly with the local mean water-table depth ($r = 0.59$) than with position, and the apparent westward deepening of 0.90 m per kilometre ($p < 0.001$) does not survive control for that depth (0.40 m per kilometre, $p = 0.081$). At 65 of the 66 reference wells the optimum lies below the mean water table, the single exception being a forest well whose drainage coefficient is not significantly positive at any datum in the sweep. That the datum is better read as a depth below ground than as an absolute elevation is confirmed by the sweep geometry: regressing each well's R^2-optimizing datum elevation on its ground elevation gives a slope of +0.868 across the 66 reference wells (mirror datum-depth slope +0.133), close to the +1 of a base at constant depth below ground and far from the 0 of a base at fixed elevation, and no single absolute elevation is definable across the network's 10.9 m spread of ground elevations. A base that follows the dune topography is consistent with the theoretical flow paths and Darcy vectors of Figure 16 and supports reading β_3 as a lumped Darcy-consistent drainage coefficient rather than requiring an independently located sea-level base. Interpolation of the per-well coefficients yields continuous surfaces whose broad structure aligns with the cluster boundaries while preserving substantial within-cluster variation (per-well ranges in Table 6; Figures 11–13). Recharge sensitivity β_1 is highest in the north-east around the Lake Edge wells (per-well up to 6.24) and declines across the open dune to its lowest values beneath the forest. Atmospheric draw β_2 peaks in the C4 forest interior and along the ridge-adjacent northern dune, and is lowest at the Lake Edge. Drainage rate β_3 shows the strongest spatial contrast of any coefficient, from the fast-draining Lake Edge (per-well up to 0.134 per month) to the near-stagnant forest interior, where one ridge-flank well (CEH14) records a negative value where the displacement formulation does not capture the boundary condition.

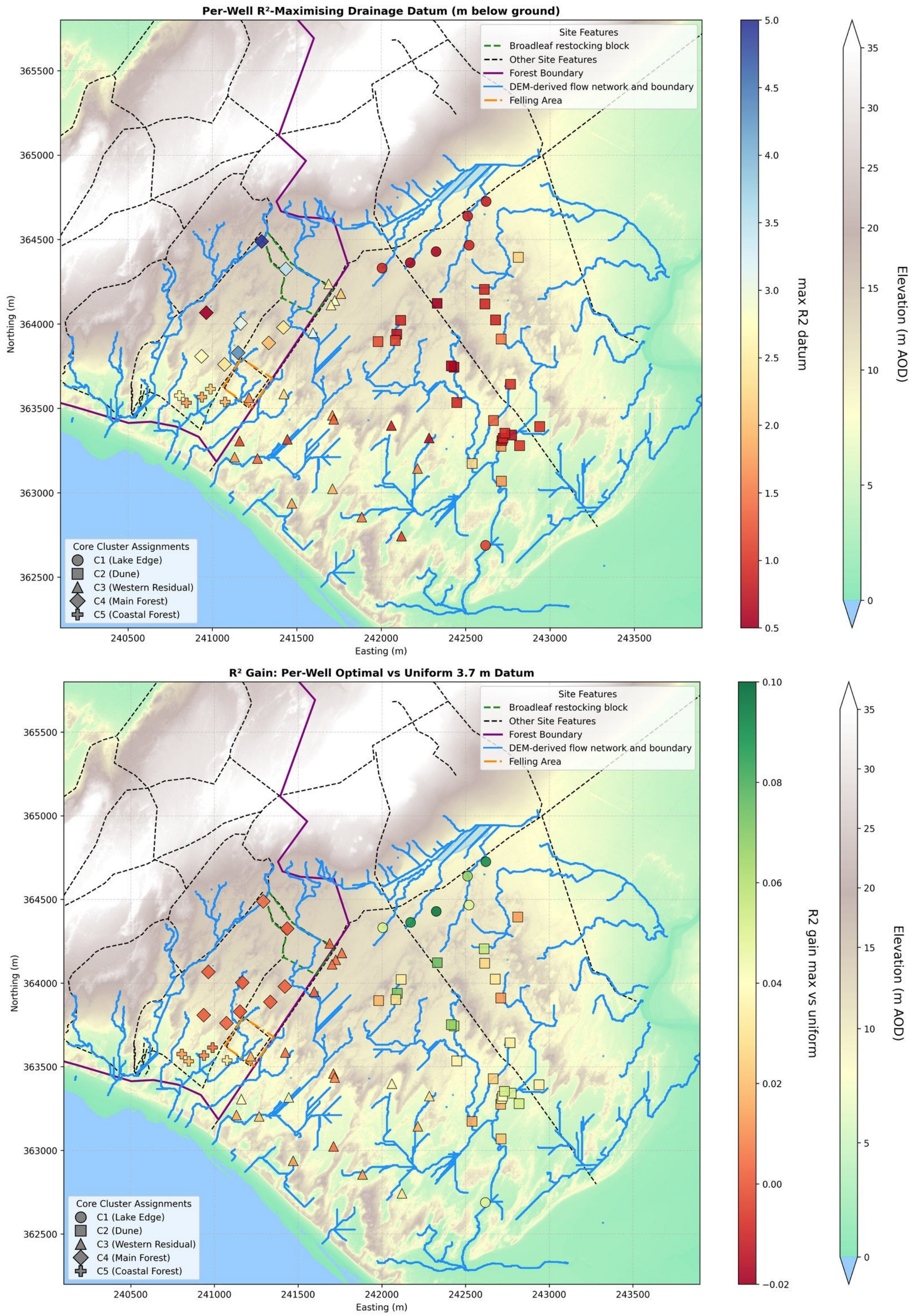

Figure 10. *Per-well optimal drainage datum (top) and the R^2 gain over the uniform 3.7 m datum (bottom).*

__Table 6.__ Per-well SSM coefficient ranges by cluster.

Cluster	β_1 range	β_2 range	β_3 range
C1 Lake Edge	4.27–6.24	−0.39–1.28	0.087–0.134
C2 Dune	3.35–5.52	1.13–2.56	0.055–0.108
C3 Western Residual	2.33–4.84	0.91–2.23	0.033–0.101
C4 Main Forest	2.07–3.48	2.04–3.83	−0.021–0.040
C5 Coastal Forest	2.06–2.75	0.80–1.38	0.038–0.057

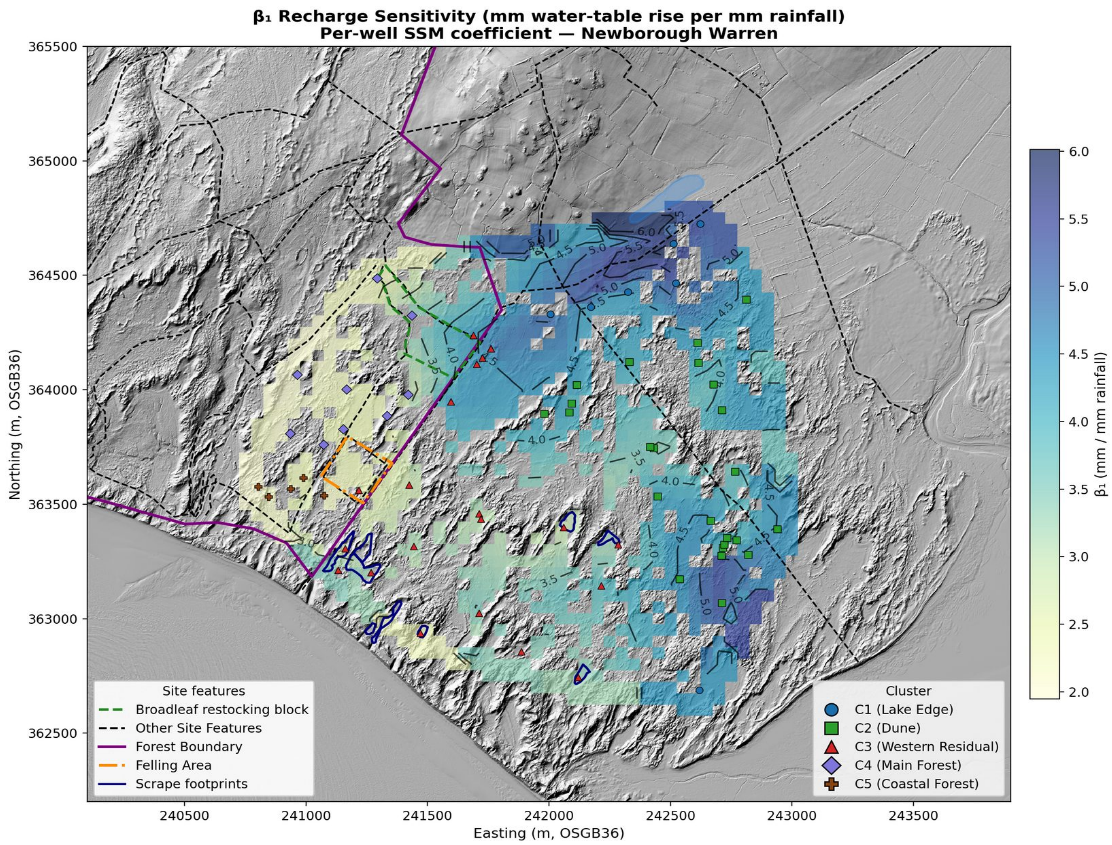

__Figure 11.__ Spatial SSM coefficient atlas for the 66-well reference network, Newborough Warren. __β_1 recharge sensitivity__ (mm water-table rise per mm rainfall) — highest at the Lake Edge (C1, 4–6 mm/mm), with a sharp drop at the forest boundary consistent with 24% canopy interception. Per-well coefficients interpolated by piecewise-linear (Delaunay) interpolation over 2 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment. (LiDAR data: © NRW & OS)

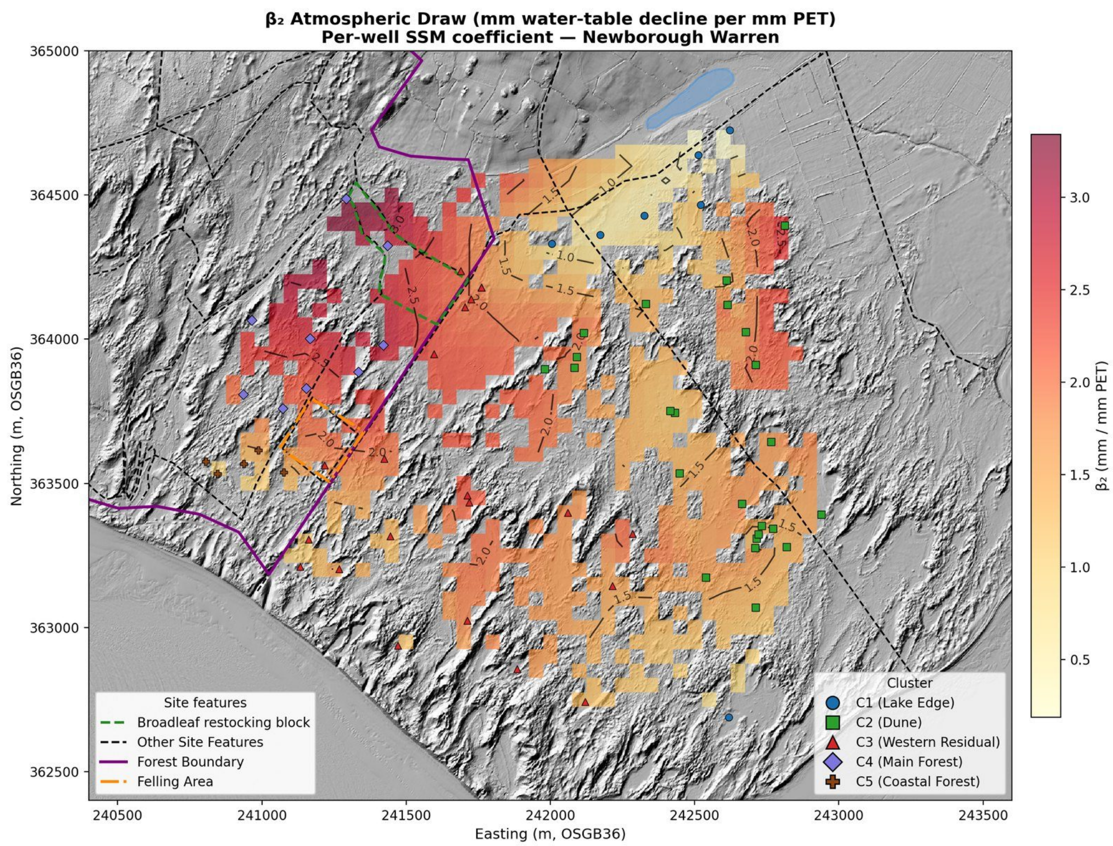

Figure 12. *Spatial SSM coefficient atlas for the 66-well reference network, Newborough Warren.* ***β_2 atmospheric draw*** *(mm water-table decline per mm PET) — highest in the forest interior (C4, 2.0–3.2 mm/mm), lowest at the Lake Edge (C1) and Coastal Forest (C5). Per-well coefficients interpolated by piecewise-linear (Delaunay) interpolation over 2 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment. (LiDAR data: © NRW & OS)*

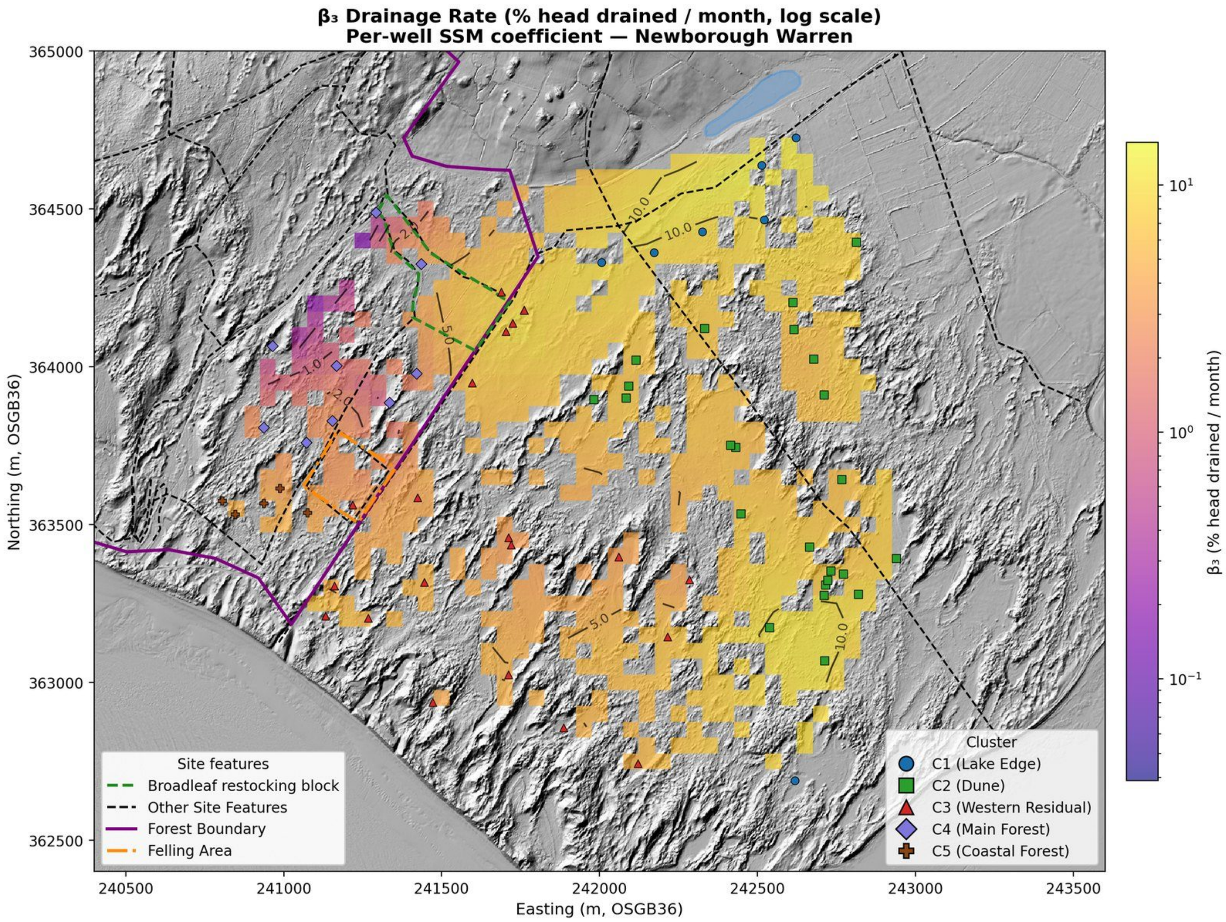

Figure 13. *Spatial SSM coefficient atlas for the 66-well reference network, Newborough Warren.* ***β_3 drainage rate*** *(% displacement drained per month, logarithmic scale) — an order-of-magnitude contrast from the fast-draining Lake Edge (7–12% per month) to the near-stagnant forest interior (C4, <3% per month). CEH14, the sole well with negative β_3, sits at the ridge flank. Per-well coefficients interpolated by piecewise-linear (Delaunay) interpolation over 2 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment. (LiDAR data: © NRW & OS)*

The drainage decay half-life $t_{½} = \ln(2)/\beta_3$ is the time for excess groundwater storage above the drainage datum to drain to half its initial value, an intuitive measure of the drainage term's recession at the project datum, depending on the drainage rate alone (Table 7; Figure 14). The half-life is the recession constant of the drainage term at the project datum — how long modelled storage above that datum takes to halve. It scales with the datum (Table 7) and is not a datum-free measure of how long the aquifer retains a perturbation. It is a timescale, not a bound on amplitude: over a summer, drainage removes only a small part of a winter excess even where the half-life is long, and the much larger seasonal fall is set by evapotranspiration (the β_2 term). The eastern open dune turns over rapidly (C1 median 7.1 months, C2 10.0), the western dune is intermediate (C3 median 13.6 months), and the forest interior drains most slowly in the network (C4 median

32.0 months, individual wells to 82). C5 Coastal Forest (median 15.8 months) resembles the western dune despite sharing the pine canopy, reinforcing the substrate and elevation control. The half-life is set by the drainage rate alone, whereas the storage-drainage index $\tau = Sy/\beta_3$ additionally weights it by specific yield; because Sy and $1/\beta_3$ are spatially independent across the network ($r = 0.11$), the two produce near-identical spatial patterns ($r = 0.98$) and the cluster ranking is unchanged. Plotted together with the SSM-over-TLM gain and specific yield as three independently derived descriptors ($t_{½}$, ΔNSE, Sy), the five clusters occupy distinct, non-overlapping regions of diagnostic space (Figure 15) — a model-based characterization of aquifer architecture consistent with, but derived independently of, the Ward's clustering.

Table 7. *Drainage decay half-life $t_{½} = \ln(2)/\beta_3$ (months) by cluster. Ridge-flank wells CEH14 (negative β_3) and CEH13 (near-zero β_3) excluded. β_3 is the recession constant of the drainage term at the 3.7 m displacement datum, so $t_{½}$ is conditional on that reference rather than a datum-free property of the aquifer. Independent recession analysis of the hydrographs places the empirical drainage base at 3.3 m, corroborating the datum, and fitting the same model on recession months alone reproduces every cluster's coefficient (ratios 1.01 to 1.24, per-well $r = +0.944$). Over ±0.5 m about the datum (3.2–4.2 m; the lower bound is set by the deepest recorded water table, 2.87 m at CEH32 in June 2012, below which the displacement changes sign) every cluster half-life rises with the reference. The cluster ranking is invariant across that range and far beyond it (C1 < C2 < C3 < C5 < C4 at every datum from 2.9 to 8.0 m). The ratios are not: relative to C3 they drift by 0.2% (C2), 2.4% (C1), 6.1% (C5) and 10.5% (C4) over 3.2–4.2 m. Absolute values and inter-cluster ratios are therefore conditional on the project datum; the ordering is not.*

Cluster	t½ min	t½ median	t½ max	n
C1 Lake Edge	5.2	7.1	8.0	7
C2 Dune	6.4	10.0	12.5	24
C3 Western Residual	6.8	13.6	21.1	21
C4 Main Forest	17.1	32.0	82.0	7
C5 Coastal Forest	12.1	15.8	18.1	5

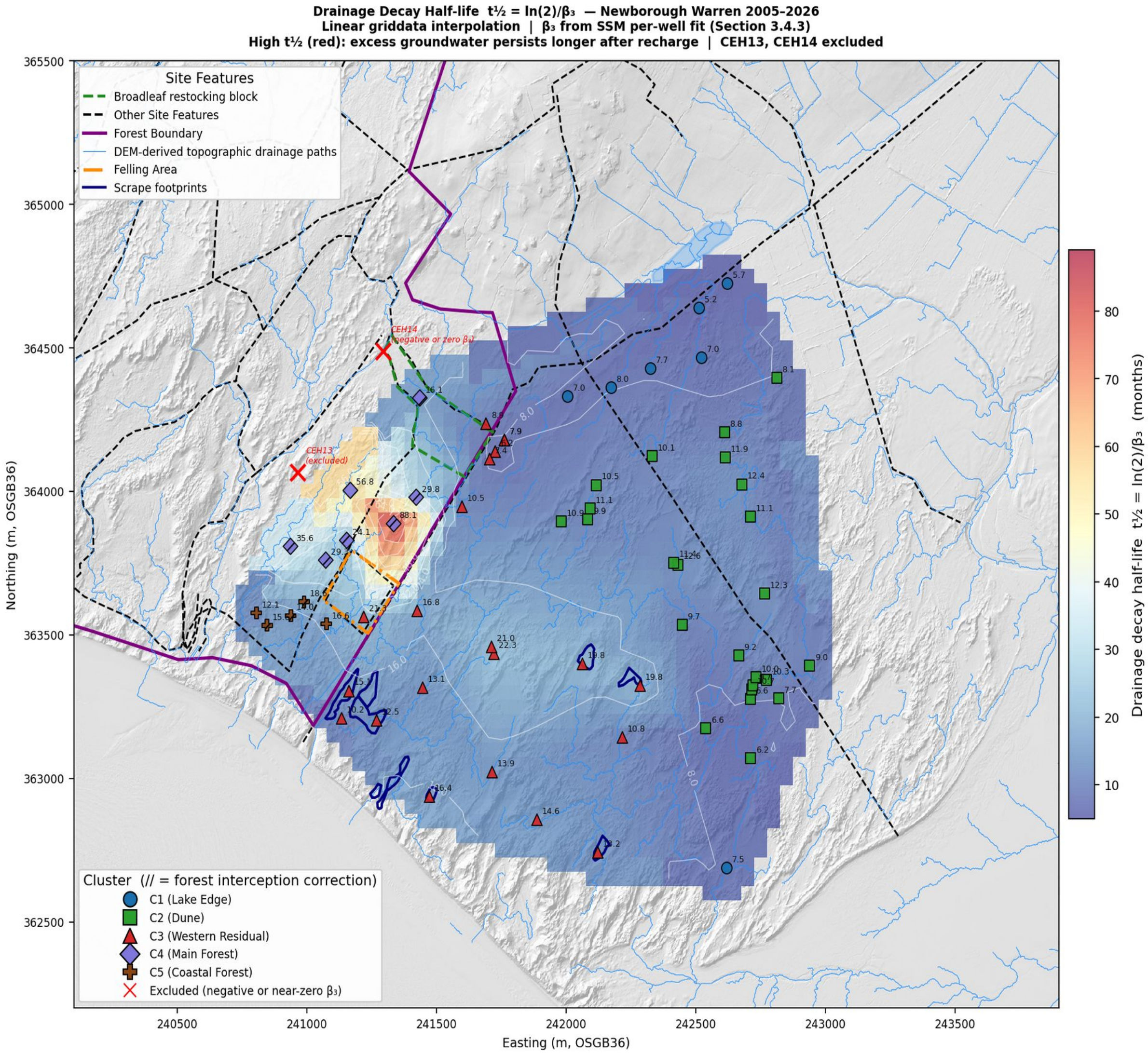

Figure 14. *Drainage decay half-life t½ = ln(2)/β₃ across the reference network. The interpolation extends over the rock-ridge bedrock outcrop on the northern boundary, which carries no monitoring wells; values shown there are extrapolations from the surrounding network and are not interpretable as model output on bedrock.*

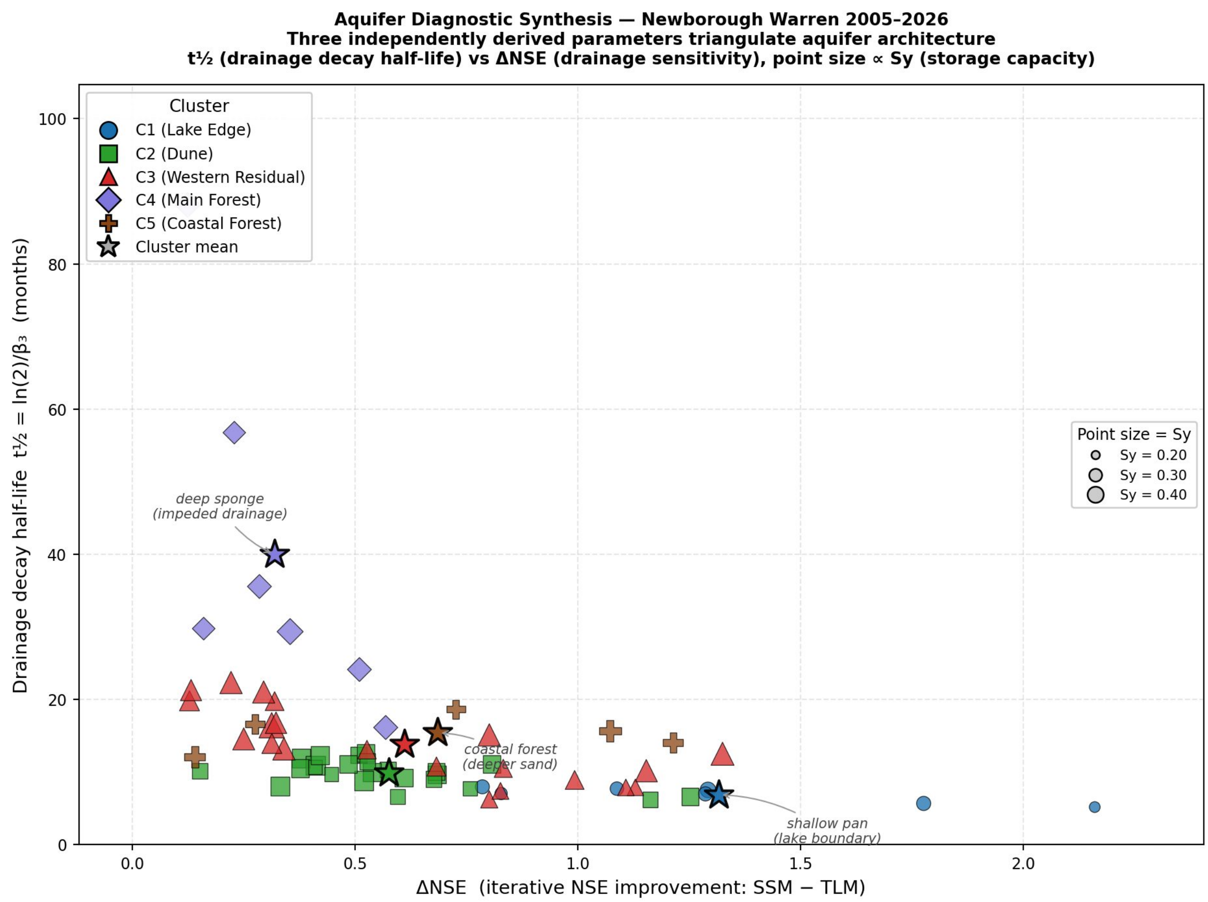

Figure 15. *Aquifer diagnostic synthesis: drainage decay half-life t½ versus SSM-over-TLM iterative NSE gain, points sized by specific yield and coloured by cluster.*

4.8 Substrate control within the forested zone

Where the canopy is uniform but the per-well coefficients span much of the site-wide range, regression against elevation, distance from the ridge and easting identifies the underlying control (Table 8). Atmospheric draw β_2 is overwhelmingly governed by elevation ($r = 0.983$, $p < 0.001$; $R^2 = 0.967$): higher wells on the ridge return systematically higher β_2. Drainage β_3 shows a weaker but significant elevation dependence ($r = -0.831$, $p < 0.001$), with distance from the ridge adding information (combined $R^2 = 0.788$). Recharge β_1 shows no significant elevation relationship ($p = 0.51$); easting is its best single predictor ($r = 0.58$, $p = 0.030$), and its correlation with distance from the ridge ($r = -0.51$) is only just short of significance ($p = 0.061$).

β_2 does vary substantially across clusters (Table 1), but the cluster-level contrast is set by substrate, water-table depth and topographic position rather than by canopy maturation: the C4–C5 split — a 2-fold difference in atmospheric draw beneath a common pine canopy — is the cleanest evidence. C4 and C5 form two distinct groups rather than a continuum. They separate almost entirely on β_2 — C4 at 2.04–3.83, C5 at 0.80–1.38, with no overlap — and on mean elevation (C4: 10.6 m, C5: 5.6 m; $p = 0.0001$), corresponding to the transition from the elevated ridge flank, with its deeper water table and unsaturated zone, to the low coastal fringe, with a shallower water table. Both zones carry the same canopy, so the separation tracks elevation and water-table depth rather than canopy. The subsurface beneath the plantation is not cored; on the topographic and coefficient evidence the C4/C5 split is best read as a substrate-and-elevation discontinuity rather than an arbitrary division of a continuous canopy population.

Table 8. *Within-forest spatial predictors of per-well SSM coefficients (n = 14). Pearson r (p).*

Coefficient	vs Elevation	vs Dist. from ridge	vs Easting
β_1 recharge	0.192 (0.512)	−0.512 (0.061)	0.579 (0.030)
β_2 atm. draw	0.983 (<0.001)	−0.905 (<0.001)	0.750 (0.002)
β_3 drainage	−0.831 (<0.001)	0.644 (0.013)	−0.480 (0.082)

4.9 Mean water-table surface and flow field

Interpolation of per-well mean elevations gives a head surface highest in the north-western forest zone adjacent to the bedrock ridge (≈14 m AOD), declining to 7–9 m AOD across the central dune and 2–3 m AOD at the southern and eastern coastal margins, with contours running approximately NE–SW (Figure 16). Normalized Darcy flow vectors computed from the head gradient diverge radially from the north-western high point: predominantly southward and south-westward in the western half of the site and south-eastward in the eastern half, broadly consistent with the DEM-derived flow network. The vectors are reported as directional context for the interpolated head field rather than as the output of a calibrated continuous-flow model.

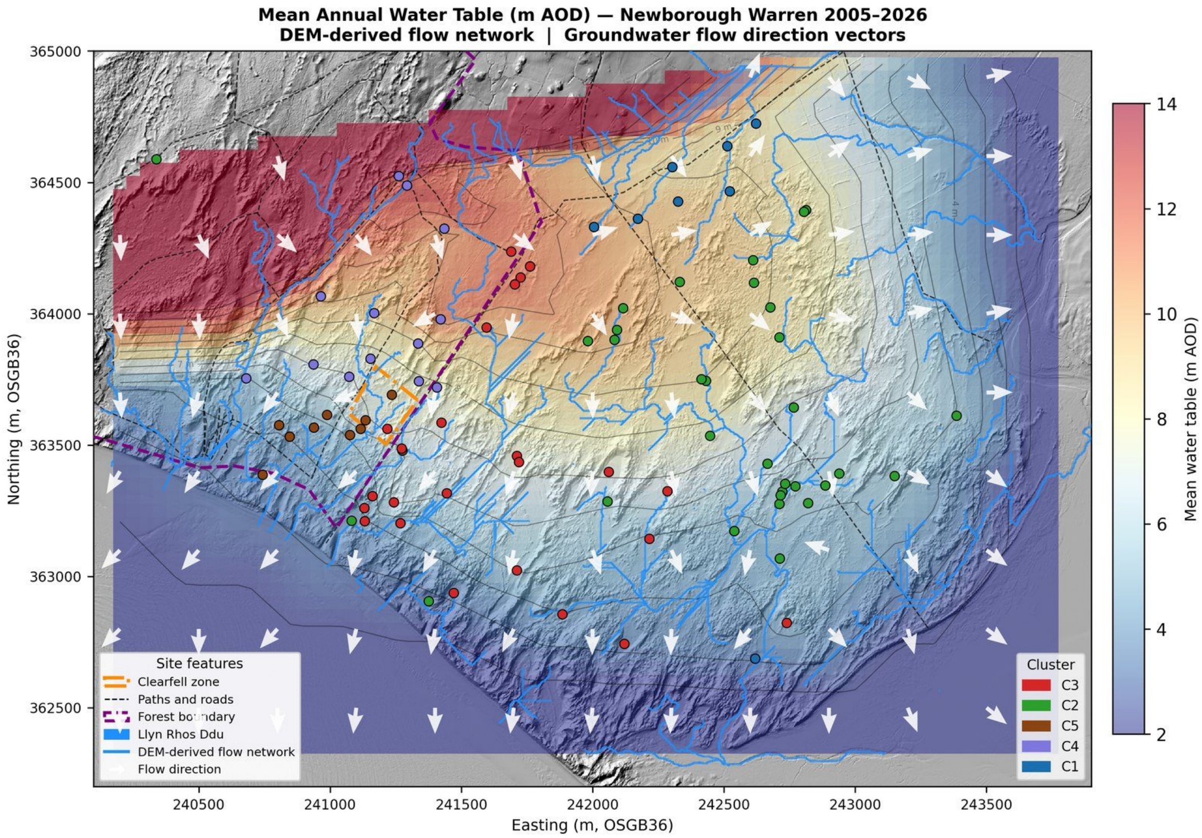

Figure 16. *Mean annual water-table surface (m AOD) with normalized Darcy flow-direction vectors. The interpolation extends over the rock-ridge bedrock outcrop on the northern boundary, which carries no monitoring wells; values shown there are extrapolations from the surrounding network and are not interpretable as model output on bedrock.*

4.10 Water-balance residual field

The residual field — modelled losses minus modelled recharge at each well, from the fitted coefficients — closes tightly and carries no coherent spatial structure, with a small systematic negative offset (Figure 17). Residuals span −0.0115 to +0.0049 m/month about a median of −0.0045 m/month, with 64 of the 66 reference wells inside ±0.01 m/month and 58 negative. All five cluster medians are negative, from −0.0065 m/month at C4 Main Forest to −0.0004 m/month at C5 Coastal Forest, and residual magnitude is uncorrelated with position on either axis (Spearman $\rho = +0.099$ against Easting, $p = 0.43$; $\rho = +0.111$ against Northing, $p = 0.37$); the signed residual carries a weak west-to-east gradient short of conventional significance ($\rho = -0.226$, $p = 0.07$), reported for completeness rather than as an established structure. No well exceeds +0.02 m/month, and only eight are positive at all: the largest are CEH4 (+0.0049), NW7 (+0.0033) and NW6 (+0.0032), in the open dune rather than at the ridge margin. No C4 Main Forest well is positive. The largest negative residual falls at the open-dune well D7 (−0.0115 m/month), with the ridge-flank well CEH14 (−0.0106 m/month) close behind. Two independent diagnostic tests on the monthly SSM residual — a cross-correlation lag test for distance-dependent transport from the ridge, and a seasonal climatology test for unmodelled summer evaporative demand — both returned null results. No unmodelled flux is required to close the balance at any cluster.

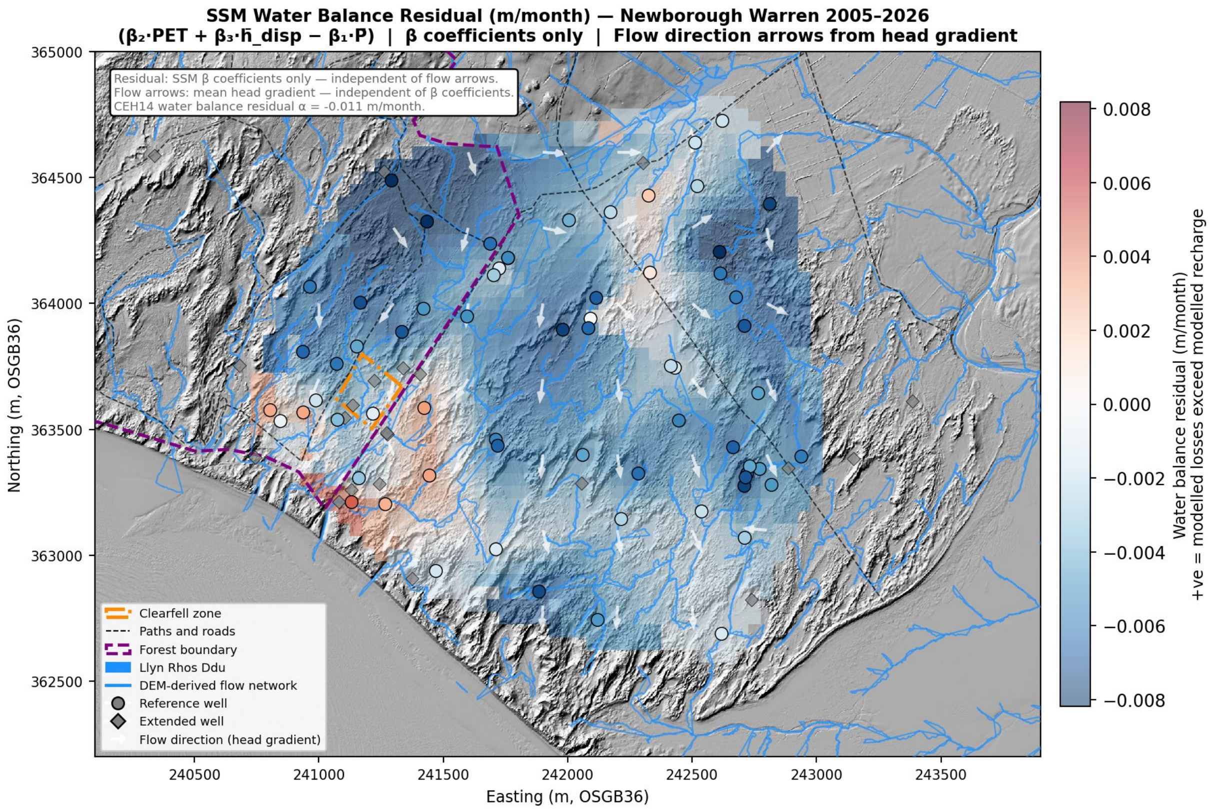

Figure 17. *SSM water-balance residual field. Positive (red) indicates modelled losses exceeding modelled recharge; negative (blue) indicates over-prediction. The field closes tightly across the network, with a small systematic negative offset: 64 of 66 reference wells lie within ±0.01 m/month and 58 are negative. No well exceeds +0.02 m/month; the largest positive value is CEH4 (+0.0049 m/month). Rainfall is gross, and climate means are taken over the same period as the mean heads. The field is a spatial diagnostic of a lumped-parameter model, not the output of a calibrated continuous-flow simulation. The interpolation extends over the rock-ridge bedrock outcrop on the northern boundary, which carries no monitoring wells; values shown there are extrapolations from the surrounding network and are not interpretable as residuals on bedrock.*

4.11 Coastal-retreat gradient and the partition of long-term decline

The network-scale regression of water-table trend against distance from the eroding Caernarfon Bay shoreline selects a linear-with-cutoff (Dupuit–Forchheimer) form in the forest-free headline specification, with a coast-edge deepening rate of −26.18 mm/yr relative to the far field, evaluated at 150 m from the shoreline (95% CI −29.0 to −23.3), an inland reach of 895 m (801–989), and a fitted far-field asymptote of −0.10 mm/yr (Figure 18). The rate is quoted at 150 m rather than at the shoreline because no well sits at the shoreline: the decay amplitude there, $\delta_0 = -31.35 \pm 1.97$ mm/yr, is an extrapolation beyond the network and is the one distance at which the two candidate decay forms disagree materially. δ_0 and L_cg are identified by the distance dependence of the trend and are unaffected by the level. The asymptote is a different quantity, and two senses of "identified" bear on it, both of which are worth stating. Statistically it is separately identified from the cumulative-water-balance covariate the panel uses to absorb spatially uniform climate forcing: the two columns are not collinear (variance inflation factor 1.01). Interpretively it is not, because a constant drift and the covariate's own trend contribution over the panel's span trade off against one another, so only their sum is recoverable. It is unstable with respect to the fitting window besides, running from −0.10 to +24.20 mm/yr as the window start is moved with the well set held fixed. On either reading it is a window statistic rather than a rate, and no far-field climate rate is read from it here (SI §S13.3–S13.4). The gradient is itself seasonal, but the boundary effect it expresses is year-round: a full-panel season × distance interaction (10,929 observations) fits the coastal term as $\delta(d)\cdot t\cdot(1 + \gamma\cdot S)$, with $S = 1$ in Mar–May, and returns $\gamma = +0.126 \pm 0.054$ ($p = 0.020$) for the linear-with-cutoff form and $+0.115 \pm 0.053$ ($p = 0.032$) for the exponential — a coastal-retreat drift rate about 12–13% steeper in Mar–May than over the rest of the year, at compatible magnitude under both decay forms. γ modulates the drift rate rather than switching it off outside spring: C5 Coastal Forest declines in both the summer minimum and the spring mean (−36 and −38 mm/yr). The all-season fit reported here is therefore a season-averaged quantity, not a season-independent one (SI §S13.6). The forest-free and full-network fits agree closely, and the C3-only specification returns a consistent coast-edge rate, indicating the gradient is not an artefact of forest-cover inclusion. Model selection by AIC favours the exponential-decay form in the forest-free specification ($\Delta AIC = -4.9$, exponential − linear-with-cutoff); the linear-with-cutoff form is adopted as the headline on physical grounds, as it embodies the Dupuit–Forchheimer expectation of a finite inland reach beyond which the migrating seaward boundary exerts no influence, and because its far-field asymptote takes the same sign as the observed far-field trend where the exponential's does not. Quoting the rate at 150 m makes that choice largely immaterial: the two forms agree to within 1.6 mm/yr there, against 9.3 mm/yr at the shoreline.

Applying the headline fit to each cluster's summer-minimum decline partitions the trend against a balanced observed basis — the slope of the annual cross-well mean of the per-well summer minima, declared in the output itself so that the basis travels with the table (Table 9). The robust result is the network curve itself — fitted across thousands of well-months of the open-dune network and insensitive to functional form — together with the C3 Western Residual decomposition, which rests on 17 open-dune wells lying within the fitted inland reach: coastal retreat accounts for about a quarter of C3's decline on that basis (mean 803 m from the coast). C1 and C2 lie beyond the fitted inland reach and carry no gradient component. C4's mean distance falls marginally inside it, at 889 m against a reach of 895 m, and carries a component of −0.21 mm/yr — negligible against its own decline, and not resolvable given a reach whose confidence interval runs from 801 to 989 m and which lengthens further under alternative climate covariates (SI §S13.4). Against balanced declines of −11.0, −11.1 and −6.5 mm/yr the modelled totals are +2.8, +2.8 and +2.6 mm/yr — the trend contribution the cumulative-water-balance covariate carries plus the fitted far-field asymptote — so the whole of their deepening, and a little more, is left unexplained rather than attributed to climate. This decomposition identifies coastal retreat as a distinct, spatially structured forcing on long-term summer minima that a climate-only attribution would fold into an apparent network-wide decline. The coastal contribution declines linearly across the reach and is zero beyond it, so it is a spatially structured forcing rather than a uniform one; the spatially uniform component is a window statistic rather than a rate, and no distance at which the two are equal follows from this fit.

An orthogonal, longitudinal test corroborates the cross-sectional gradient. Along a shore-normal transect from a coastal anchor (CEH22) to an inland anchor beyond the fitted reach (NW4), the spring (MAM) coast-to-inland head difference deepens at −28.2 mm/yr (bootstrap 95% CI −34.2 to −22.0; AR(1)-corrected; 2010–2023), consistent with the modelled coast-edge rate of −26.18 mm/yr. The coastal anchor sits at 147 m, so the comparison is made at essentially the distance at which the modelled rate is quoted. Referencing the coastal record to the erosion-free inland anchor removes common-mode climate without a model, and the coastal-end drawdown then orders monotonically through time (Spearman ρ improving from −0.32 for the raw record to −0.87 after referencing; Figure 18c). A coast-to-inland difference that grows through time is the signature of a migrating seaward boundary rather than a static geometric offset or a climate wobble, and provides an observational counterpart to the cross-sectional fit on the same record.

The C5 Coastal Forest row is reported with a caveat rather than as a firmly quantified cluster attribution. After the clearfell-zone exclusion and the requirement for a valid summer-minimum slope, three C5 wells enter the gradient subset — ceh3 at 176 m from the coast, nw9 at 419 m and nw8 at 521 m, mean 372 m — declining at −32.0, −32.7 and −59.2 mm/yr respectively. All three are statistically significant ($p = 0.0002$, 0.002 and 0.005), and together they are much the steepest group in the network. The gradient attributes −18.3 mm/yr across them, which is 113% of the cluster's balanced basis of −16.1 mm/yr, leaving −0.7 mm/yr unexplained. That the attribution slightly exceeds the basis reflects how weakly three wells constrain that basis rather than any failure of the curve: the same three wells give a per-well mean of −41.3 mm/yr and a Script-14 centroid of −35.9 mm/yr, so the choice of basis moves the denominator by more than a factor of two while the numerator is fixed by the fit. The percentage should therefore be read as indicating that coastal retreat is sufficient to account for C5's decline, not as a calibrated share. Because the forest-free curve is fitted on the open-dune wells and excludes the forested clusters entirely, none of the three can bias the fit, and the choice of functional form does not bear on the reading: across their distance range the linear-with-cutoff form predicts deepening from −25 mm/yr at ceh3 to −13 mm/yr at nw8, and the exponential is of the same order throughout. The cluster is therefore treated as corroborating the gradient's direction and magnitude rather than as an independent cluster-level estimate, and is cross-referenced to the substrate and coastal-position interpretation of C5 in Section 4.8.

Table 9. *Decomposition of cluster summer-minimum decline (mm/yr) against a balanced observed basis. The observed column is the slope of the annual cross-well mean of the per-well summer minima over the gradient-analysis subset — one regression on one annual series, rather than an average of per-well fits taken over unequal record windows, which reports a composition of record lengths in place of a rate (SI §S13). The climate + offset column is the trend contribution the cumulative-water-balance covariate carries over the fitted span plus the fitted far-field offset c_far; the asymptote is a window statistic rather than a rate, so this column is not a measured climate background. Well counts here are the gradient-analysis subset (clearfell-zone wells dropped; wells require a valid summer-minimum slope), not the cluster sizes of Table 1. Components may not sum exactly to the observed basis because of rounding.*

Cluster	mean dist. (m)	observed	coastal-retreat	climate + offset	unexplained	coastal %
C1 Lake Edge	1819	−11.0	0	+2.8	−13.8	0
C2 Dune	1473	−11.1	0	+2.8	−13.9	0
C3 Western Residual	803	−14.2	−3.2	+2.8	−13.8	23
C4 Main Forest (n = 4)	889	−6.5	−0.2	+2.8	−9.1	3
C5 Coastal Forest (n = 3) †	372	−16.1	−18.3	+2.8	−0.7	113 †

† C5 rests on three open wells in the gradient subset and lies closest to the shore, so the fitted gradient slightly over-explains its decline: a share above 100 % is possible because each component is a rate rather than a share of a fitted total. Reported for completeness, not as a quantified attribution (see text). The Script-14 cluster-centroid trend, retained in the output as a context column, is −35.9 mm/yr against a balanced basis of −16.1 mm/yr.

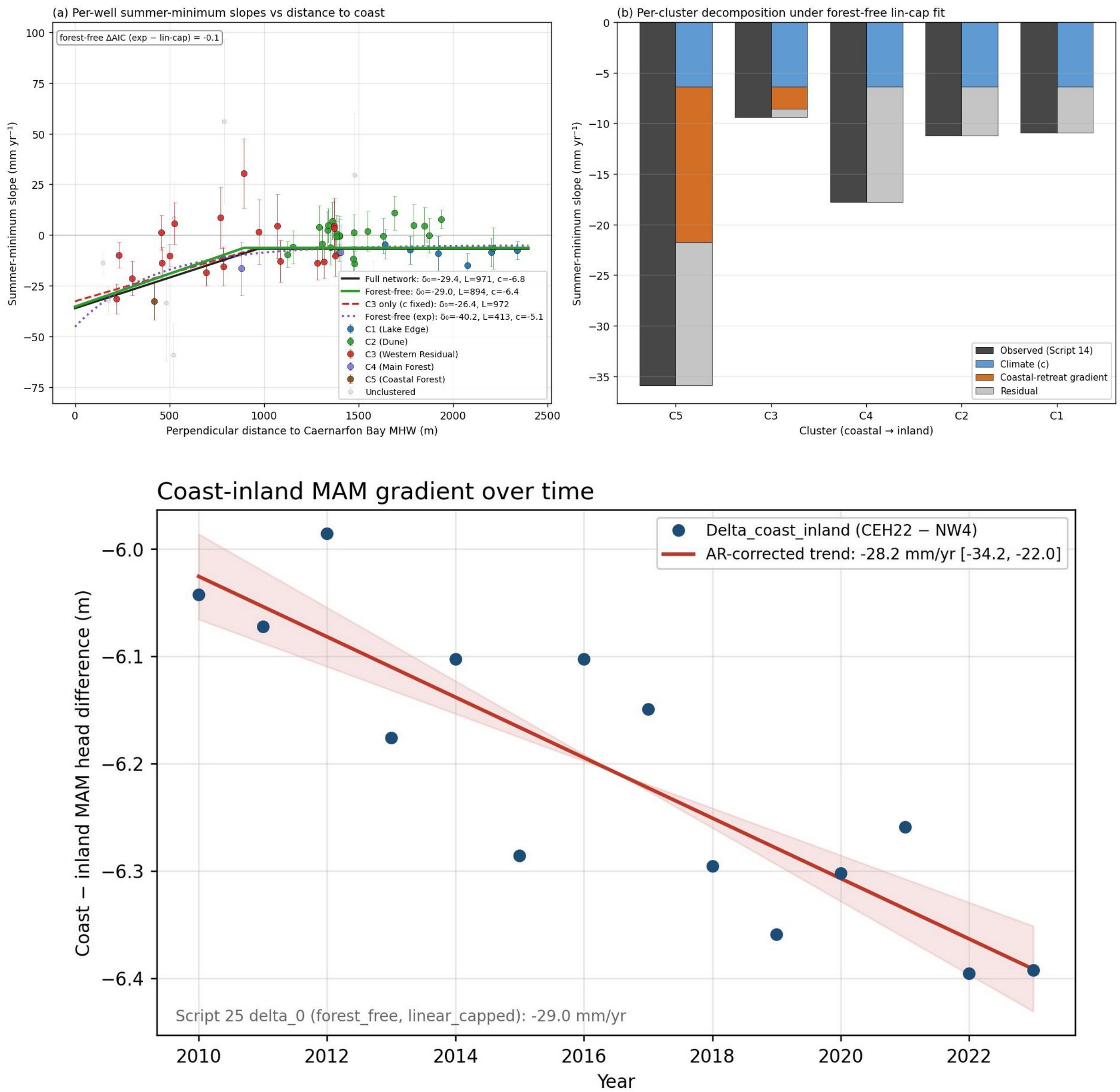

Figure 18. *Coastal-retreat gradient. (a) Per-well summer-minimum slope versus distance to the eroding Caernarfon Bay shoreline, with the full-network, forest-free and C3-only linear-with-cutoff fits and the forest-free exponential fit overlaid; b) per-cluster decomposition of the observed decline into climate, coastal-retreat gradient and residual; (c) observational corroboration — the spring (MAM) coast-to-inland head difference (coastal anchor CEH22 minus inland anchor NW4) against year, with the AR(1)-corrected trend of −28.2 mm/yr; this transect subset is distinct from the whole-network fit in (a) and (b).*

5. Discussion

5.1 Geology, land cover and the cluster structure

The five hydrogeological clusters resolved by hierarchical clustering of hydrograph dynamics map onto the documented geological structure of the site. The broad contrast between the responsive eastern open dune (C1 Lake Edge, C2 Dune) and the buffered south-western dune (C3) reflects the difference in subsurface stratigraphy described by Stratford et al. (2007): shallow sand over till and estuarine deposits in the east constrains storage and drives flashy water-table responses, while a deeper sand column in the south-west supports a more buffered aquifer. The mechanistic coefficients recover this divide independently of the clustering that defined it, and they resolve a gradient running through the open dune rather than a simple binary. C1 Lake Edge is one extreme — the highest recharge sensitivity in the network (β_1 = 4.58) with the fastest head-dependent drainage ($-\beta_3$ = 0.089) and the fastest turnover in the network (t½ ≈ 7.1 months), a fill-and-drain regime in which a shallow base promotes rapid lateral exchange directly with Llyn Rhos-Ddu. C2 Dune, the largest and most representative unit (24 wells), is the open-dune archetype between the end-members: a still-high recharge sensitivity (β_1 = 3.97) and a drainage-dominated loss partition (68% of losses), but a slower drainage decay ($-\beta_3$ = 0.064) and a longer turnover (t½ ≈ 10.0 months) than the lake-coupled C1, reflecting open dune without a direct lake connection. C3 Western Residual is the buffered opposite — the lowest recharge sensitivity of the open clusters (β_1 = 3.57), the slowest open-system drainage ($-\beta_3$ = 0.057) and the slowest open-dune drainage (t½ ≈ 13.6 months at the project datum) — as the deep aeolian sand attenuates both the rainfall pulse and its release. The forested interior fills slowly and drains most sluggishly of all. C4 Main Forest is the cluster geographically nearest the rock ridge, with the metamorphic bedrock outcrop forming its northern margin — a setting reflected in the impeded drainage and the elevated atmospheric-draw coefficient (Section 5.2) and revisited in the residual-field discussion (Section 5.4).

This gradient also bears on how C3 is best understood. Its membership is the most diffuse in the network — the great majority of its wells are gradationally assigned, correlating almost as strongly with the C2 Dune centroid and with the coastal forest as with their own — and it is the open-dune cluster closest to both the plantation margin and the eroding Caernarfon Bay shoreline. Rather than a discrete western block, C3 is therefore better read as the transitional, more strongly buffered south-western continuation of the open dune, with a coherent central signature (the longest open-dune turnover and the lowest open recharge sensitivity) but boundaries that grade into both the dune and the forest. One parsimonious mechanism, consistent with the surface geology but not directly observed here, is a single estuarine-clay base that crops out on the higher eastern and southern ground — where a thin overlying sand column produces the flashy C2 response — and dips beneath the surface toward Caernarfon Bay, where a thickening sand column buffers the C3 response, making C2 and C3 one dipping system rather than two substrate blocks. The static buffering gradient is consistent with this: the drainage half-life lengthens from roughly 10 months in C2 to about 13 months in C3 as the coast is approached (Table 7), and the drainage coefficient slows in step (Table 1) — the opposite of the flashier response a discharge boundary alone would produce. The storage evidence points the same way and is independent of the coefficients: within the open dune the water-table-fluctuation specific yields rise on a coherent planar trend toward the south-west (Section 4.5), the direction in which a dipping base would thicken the sand column. Because coastal proximity, the plantation margin and the inferred clay dip all covary along the same south-western axis, this evidence supports the transitional reading without isolating the clay dip as its sole cause; a cored or geophysical transect would be needed to test it directly (Section 5.7). The two reaches responsible — the forest-interception drawdown propagating inland from the plantation, and the coastal-retreat drawdown extending inland from the eroding front — are mapped in Figures 19 and 20, which place the C3 wells within both reaches while the C2 wells lie largely beyond them.

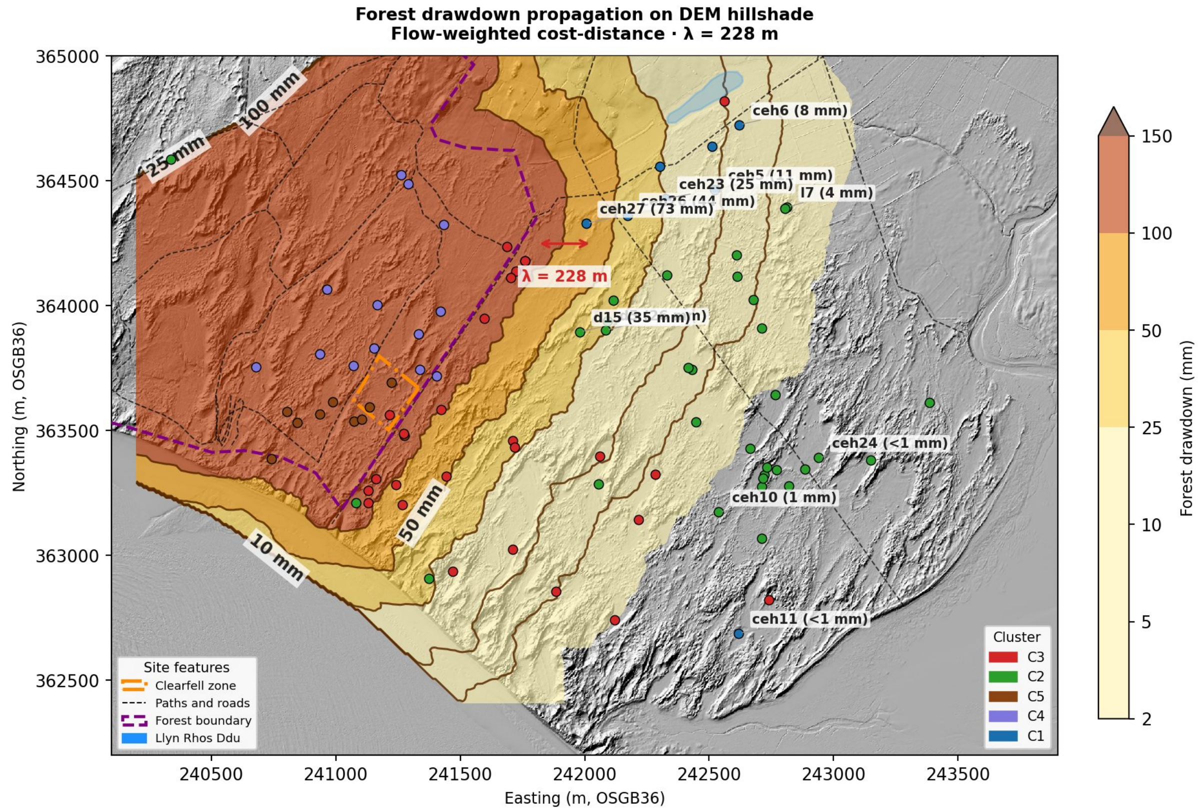

Figure 19. *Illustrative forest reach. Forest-interception drawdown propagated inland from the plantation boundary by flow-weighted cost-distance (decay length λ ≈ 230 m, from C3's specific yield and drainage coefficient combined with an assumed hydraulic diffusivity — a literature conductivity, K = 6 m/day (Betson et al., 2002), and a nominal 5 m saturated thickness, the latter poorly constrained and plausibly larger; assumed edge deficit). The reach is a scaling estimate rather than a calibrated length: across the range reported for dune sand (K = 2–20 m/day) λ spans roughly 130–420 m, and across a 3–8 m saturated thickness 180–290 m, so it is stated here to the nearest tens of metres. The mapped pattern depends only on the reach being of order 200 m rather than of order 20 m or 2 km, which holds across that whole range, over the DEM hillshade with per-well drawdown values labelled. The C3 wells fall within the forest reach while the C2 wells lie largely beyond it — drawdown falls from tens of millimetres near the plantation to under 1 mm at the easternmost C2 wells; the forest boundary, clearfell zone and Llyn Rhos-Ddu are marked. Drawdown magnitudes are illustrative of the reach rather than measured values.*

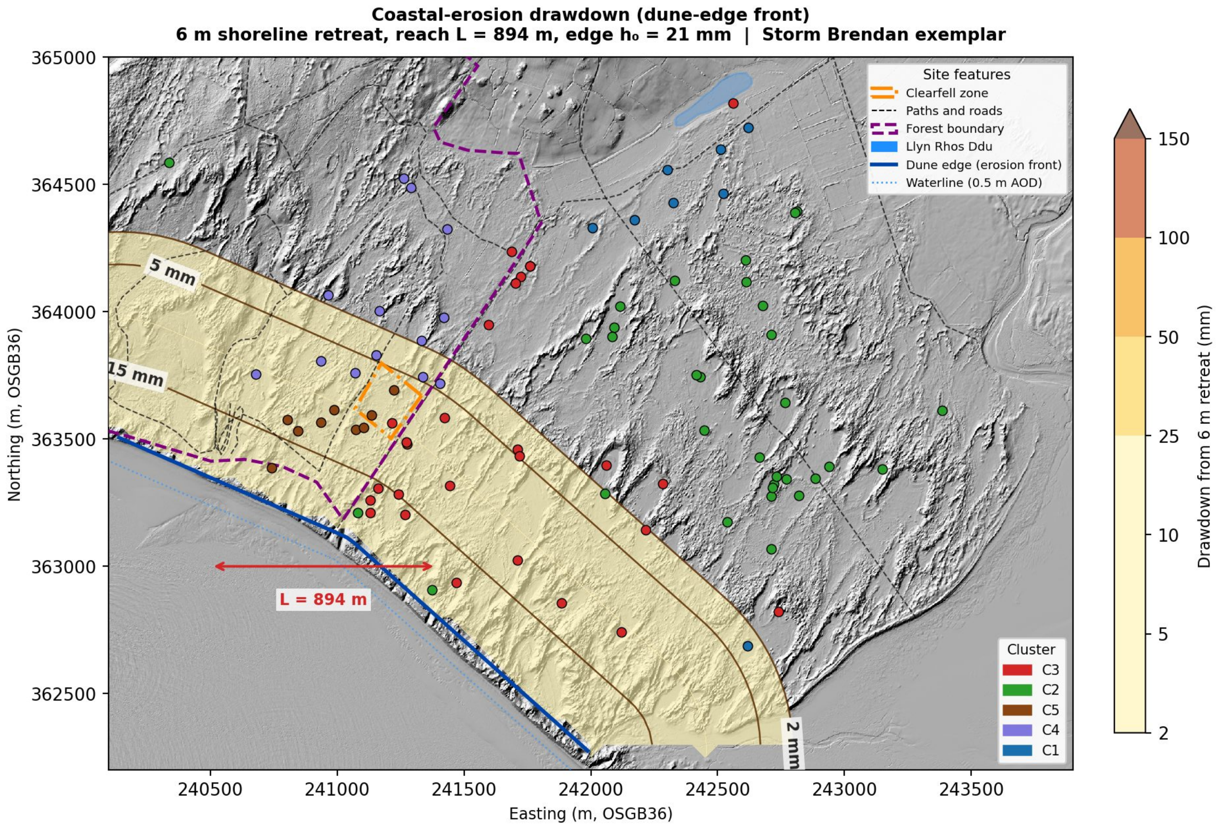

Figure 20. *Episodic coastal-retreat reach. Water-table drawdown contoured inland from the eroding dune-edge front for a single Storm Brendan-scale retreat event (≈6 m at the front, early 2020), over the fitted inland reach (L_cg = 895 m; §4.11). Coastal retreat at this shore is episodic, concentrated in storm events; the per-event edge drawdown is scaled from the chronic gradient through the retreat rate measured over the same period as that gradient (E_c = 2.32 m/yr, 2006–2026, from shoreline positions digitized in this study). The C3 wells fall within the reach while the C2 wells lie largely east of it; the forest boundary and Llyn Rhos-Ddu are marked.*

That the clustering (Figures 4 and 5), the mechanistic coefficients (Figures 11–13 and Table 1), and a third, fully independent diagnostic synthesis (Figure 15) all converge on the same structure is the strongest evidence that the partition is real rather than an artefact of the linkage method. When the drainage decay half-life ($t½ = \ln(2)/\beta_3$; Figure 14), the SSM forecasting gain over the linear benchmark (ΔNSE; Figure 9), and the water-table-fluctuation specific yield (Figure 8) — three parameters derived by separate routes — are plotted together (Figure 15), the five clusters occupy distinct, non-overlapping regions of diagnostic space. C1 Lake Edge sits at the shallow, fast-cycling, drainage-dominated extreme; C4 Main Forest at the sluggish, atmospheric-draw-dominated opposite; and C5 separates cleanly from C4 despite sharing the same canopy. This is a model-based characterization of aquifer architecture that is consistent with, but independent of, the Ward's clustering applied to the raw hydrographs.

5.2 Substrate versus surface control: interpreting the coefficients

The two forested clusters are inferred to share the deep sandy substrate of the western block, and carry the same Corsican pine canopy, yet segregate into distinct units — and resolving why is central to interpreting what the SSM coefficients measure. Within the forest zone, elevation alone explains 97% of the variance in atmospheric draw (β_2 versus elevation, $r = 0.983$), with C4 and C5 separating completely on this axis and on mean elevation (10.6 m versus 5.6 m). C4, on the elevated ridge flank, has a deeper water table and unsaturated zone — which we interpret as permitting stronger atmospheric draw — together with weaker drainage; C5, on the lower coastal fringe, has a shallower water table and faster drainage. The measured contrast in water-table depth and coefficient behaviour is unambiguous. That interpretation has independent measured support at this site. Freeman (2008) recorded the water table beneath the Newborough plantation approximately 1.07 m below the open Warren (ANOVA $p < 0.001$) and mean volumetric soil moisture at roughly half the open-dune value, attributing the moisture contrast to the loss of capillary contact between the upper soil layers and the water table where the water table is deeper. The forest zone therefore carries an unsaturated zone that is both thicker and drier, consistent with the coefficient contrast we recover. The substrate explanation differs in confidence between the two clusters. For C4, the bedrock of the ridge outcrops within the forest, so the nature of the rock underlying the cluster is established; its depth and extent beneath C4 are not known without coring, but the impeded drainage there is consistent with a shallow, irregular bedrock base close beneath the wells — an inference supported by the proximity of the outcropping ridge. The storage evidence within the plantation supports it directly. Across the fourteen forested wells the specific yield falls with elevation ($r = -0.95$, $p < 0.001$) and rises with distance from the ridge ($r = +0.84$, $p < 0.001$) — the pattern expected where a shallowing bedrock base thins the drainable sand column toward the outcrop. The relationship is robust to the exclusion of the two anomalous ridge-flank wells ($r = +0.83$ against ridge distance with both removed) and is present in the uncorrected estimates as well as the corrected ones ($r = +0.67$, $p = 0.008$), so it is not an artefact of the canopy correction. For C5, the proposed more-uniform, deeper coastal sand is the weaker inference, resting on the coefficients and topography alone. On that basis the split is best read as a substrate-and-topographic-position discontinuity beneath a constant canopy rather than an arbitrary division of a continuous population. The forest-interception drawdown footprint (Figure 19) occupies the same corner, so the high atmospheric draw, the positive water-balance residual and the modelled drawdown reach together describe one coherent forest signature; because all three are anchored to the same plantation-and-ridge zone, however, their spatial agreement is mutual consistency rather than independent corroboration.

This finding clarifies a potential misreading of the cluster contrasts. The forested clusters carry the highest LCSC values in the network, which might be read as low aquifer drainability. But LCSC is a recharge-sensitivity descriptor, not a storage parameter: the fitted β_1 absorbs canopy interception alongside the true storage response, so part of the forest clusters' elevated LCSC reflects the suppression of throughfall by the canopy rather than a difference in the aquifer itself. The water-table-fluctuation specific yields settle this directly. The uncorrected forest medians exceed the open-dune range, because the method attributes the full rainfall total to recharge when in reality the canopy intercepts roughly a quarter of it; once corrected for interception, C4's forest median (0.260) falls within the open-dune range (0.210–0.327) and C5's (0.321) sits at its upper edge, though C5's estimate is constrained by the physical-plausibility limit and is only weakly corroborative. The interpolated specific-yield surface shows no discontinuity at the cluster margins. The effective storage inferred from the water-table response therefore varies smoothly and continuously across the site, stepping nowhere at the cluster margins — consistent with, though on these data not proof of, a well-mixed aeolian depositional history — with the C4 specific yield and the continuous Sy surface, rather than the ceiling-bound C5 estimate, carrying the argument that the forest clusters' distinctive coefficient signature derives largely from a surface boundary condition (canopy interception) rather than from a different subsurface architecture.

A depth-dependent extension of the model, in which atmospheric draw declines as the capillary fringe retreats from the root zone with increasing water-table depth, confirms that a real depth-coupling effect is present at every cluster, strongest where the water table is shallowest (C1) and weakest where it is deepest (C4). Notably, C4 shows the weakest depth coupling despite carrying the highest atmospheric-draw coefficient, confirming that its high β_2 reflects low specific yield rather than a depth-dependent evapotranspiration pathway. The standard fixed-β_2 model absorbs the coupling as an effective depth-averaged parameter. The depth coupling and the lateral-drainage geometry together contribute to the within-forest β_2 variation alongside substrate, but the dominant control on the cluster-level contrast remains topographic position and substrate rather than canopy. These within- and cross-cluster comparisons rest on the ordering of β_2 rather than on its absolute magnitude: the fitted β_2 values are OLS estimates biased somewhat high in absolute terms, and are reliable as rankings rather than as quantitative atmospheric-draw rates.

5.3 Recharge, loss and the water balance

Decomposed under identical climate forcing, the clusters reveal markedly different recharge sensitivities and loss partitions while closing the monthly water balance to within 1.8% of total losses at every cluster. Recharge declines systematically from the Lake Edge through the open dune to the forested clusters, and total losses track recharge closely, but the partition between drainage and atmospheric draw varies sharply: drainage dominates at C1 (85% of losses), reflecting rapid lateral exchange with the lake, while atmospheric draw dominates at C4 (76%), where the suppressed recharge pulse never lifts the water table far enough above equilibrium to generate a strong drainage response and the irregular substrate impedes what drainage there is. The suppression has two components. Interception removes a fraction of rainfall at the crown, and the profile beneath the canopy is measurably drier — Freeman (2008) recorded roughly half the open-dune volumetric soil moisture beneath the plantation — so a larger storage deficit must be satisfied before the remainder reaches the water table. Her felled clearing plot, which carried a water table statistically indistinguishable from the open Warren, indicates that the second component is a consequence of the canopy rather than an independent property of the forest zone. The displacement formulation is what makes this contrast resolvable: under a ground-surface reference the C4 drainage term is borderline, but referencing drainage to a fixed sub-surface datum recovers a clearly significant coefficient at all five clusters, correctly specifying the hydraulic level against which drainage operates.

Two cautions attach to the recharge term. First, because the model is fitted without an intercept, in an open-boundary system the recharge coefficient can absorb unmeasured boundary fluxes (Young, 2011; Knotters and van Walsum, 1997); $\beta_1 \cdot P$ should not be read as a direct measure of local precipitation reaching the water table. Second, the atmospheric-draw coefficient is an effective parameter: Thornthwaite PET is an energy-based demand index that does not define a reference surface in the manner of Penman–Monteith, so β_2 folds together transpiration, interception, specific yield and depth coupling into a single scaling between atmospheric demand and water-table response, and a formal crop-coefficient decomposition does not apply. Both cautions are consistent with the substrate interpretation above: the coefficients are diagnostic of behaviour, and their physical decomposition requires the independent specific-yield and depth-coupling evidence assembled here.

5.4 Boundary conditions and the water-balance residual field

The near-closure of the water balance at the cluster mean also holds in space. The residual field — modelled losses minus modelled recharge, from the fitted coefficients — lies within ±0.01 m/month at 60 of the 66 reference wells, carries no gradient across the site, and returns cluster medians spanning −0.005 to +0.005 m/month. A positive residual would mark a location where measured rainfall-driven recharge does not fully account for the observed water level, implying an unmodelled input; three open-dune wells carry such a residual above +0.02 m/month, two of them effectively co-located, and no forest or ridge-proximal well does. The largest negative residual falls at the ridge-flank well CEH14. On this evidence no ridge-derived lateral input is required to close the balance, and the residual field provides no positive support for one.

Two further diagnostics constrain what the record could have detected had such an input been present. Two independent diagnostic tests — a cross-correlation lag analysis for distance-dependent transport from the ridge, and a seasonal climatology analysis for unmodelled summer evaporative demand — both returned null results. The cross-correlation null is itself qualified by sampling frequency: monthly observations are an order of magnitude coarser than the days-to-weeks transit times expected for fracture-flow input from a metamorphic bedrock ridge, so a real ridge-derived lag could be invisible at this resolution, and the null result bears on what the present record can resolve rather than on whether the mechanism is operative. The seasonal phase of the residual narrows the field further: it peaks in winter and early spring at the great majority of wells and at none in summer, which is inconsistent with any vertical-flux parameterization error — systematically underestimated summer PET, an incomplete canopy-interception correction, or any under-resolved summer evaporative demand, all of which would peak in summer when those fluxes are active — and consistent with either a winter-phased nonlinear recharge mechanism or ridge-derived lateral input arriving during the months of greatest rainfall. A cluster-stratified version of the same climatology shows mean winter-minus-summer residual contrasts of +7.4 mm in C3 ($p<0.001$), +6.2 mm in C4 ($p=0.086$, marginal) and −2.8 mm in C5 (not significant); the C3 > C4 > C5 ordering and the magnitude of the C3 signal together rule out canopy-interception over-estimation as the sole driver of the winter–summer contrast, since the largest signal sits in a cluster with no canopy. The residual pattern is real and reproducible; its mechanistic attribution is unresolved. This is the appropriate stance for a lumped-parameter framework: the residual field is a structural diagnostic of where the three-term balance is insufficient, not a quantified map of lateral flux, because it also absorbs nonlinear soil-moisture dynamics, coefficient sensitivity and any systematic PET bias.

5.5 Long-term forcing: coastal retreat and the partition of decline

The network-scale gradient regression isolates a forcing that a climate-only analysis would misattribute. Fitted across the open-dune network, the regression resolves a coast-edge water-table deepening of roughly 26 mm/yr at 150 m from the shoreline, declining to zero over an inland reach of about 900 m, and of at most about 1 km under alternative climate covariates (SI §S13.4). What that decomposition licenses is the removal of the coastal term, not the recovery of a climate rate. The fitted far-field asymptote is a window statistic rather than a rate — it moves across a range of some 24 mm/yr as the fitting window is varied with the well set held fixed — so no clean climate baseline follows from the fit, and none is offered here. An attribution analysis at this site must estimate its own baseline; what it should take from the decomposition is that a spatially structured coastal forcing exists at the western shore and has to be separated first, rather than folded into an apparent network-wide decline. The result is robust to functional form at the network scale and to forest-cover inclusion, and it identifies coastal retreat as a spatially structured forcing distinct from climate: only the near-coast western clusters carry a gradient component, with coastal retreat accounting for about a quarter of the observed summer-minimum decline at C3, while the inland clusters carry either no gradient component or, at C4, one too small to resolve.

The aquifer's drainage decay half-life $t_{1/2} = \ln(2)/\beta_3$, of order 7–32 months across the clusters (Table 7), exceeds the duration of individual storm-driven retreat events by an order of magnitude, so that the aquifer functions as a low-pass filter on the shoreline forcing — smoothing discrete event-scale shocks into a chronic gradient and validating the steady-state Dupuit–Forchheimer regression as the appropriate kinematic form. The chronic gradient resolved here is the time-averaged expression of a retreat process that, at this shore, is episodic rather than steady (Pye & Blott, 2024): shoreline loss is concentrated in storm events, of which the early-2020 Storm Brendan — a discrete retreat of order several metres at the dune front — is a recent example, set against a mean of E_c = 2.32 m/yr measured here from shoreline positions digitized across 2006–2026, which is the rate used to scale the gradient because it spans the period over which the gradient is fitted. Reported rates for this frontage vary with window, extent and method: Pye & Blott (2024) record up to 16.3 m at the southern Twyni Penrhos profile over 2013–2022, about 1.8 m/yr at the most active point, and a contemporary account of the 2014–

2020 window reports roughly 50 m of loss concentrated in storm events (Forgrave, 2020). Figure 20 maps the inland drawdown reach implied by a single Brendan-scale event; the steady-state regression averages many such events into the per-year deepening rate, so the gradient and the episodic map are two views of the same forcing rather than competing estimates. Consistent with this, the coast-to-inland head difference is observed to deepen through the record at −28.2 mm/yr once common-mode climate is removed model-free (Section 4.11): a growing gradient is the temporal signature the steady-state regression predicts, distinguishing a migrating boundary from a static offset, though it cannot on its own separate erosion from a slowly varying substrate — the cored shore-normal transect (Section 5.7) remains the mechanism-resolving test. Figure 21 draws the coastal forcing along that reach, largest at the shore and declining to nothing across the ~900 m span. The fitted far-field constant is a window statistic, not a site-wide climate background, and is not drawn as a curve. The equilibration time for a boundary perturbation to propagate across this reach, set by the same hydraulic diffusivity, is of order years to decades (Gerla, 2019), so the coastal signal registers as a chronic, quasi-steady gradient rather than a response to any single retreat event. A rising sea level, moreover, raises the water table at the coast and decays inland over a comparable reach, so the fitted coast-edge gradient is the net of two opposing coastal processes: sea-level rise of approximately 4 mm/yr (UKCP18 north-Wales near-term central estimate) works against the erosion-driven drawdown, making the fitted δ_0 a conservative estimate of the erosion component.

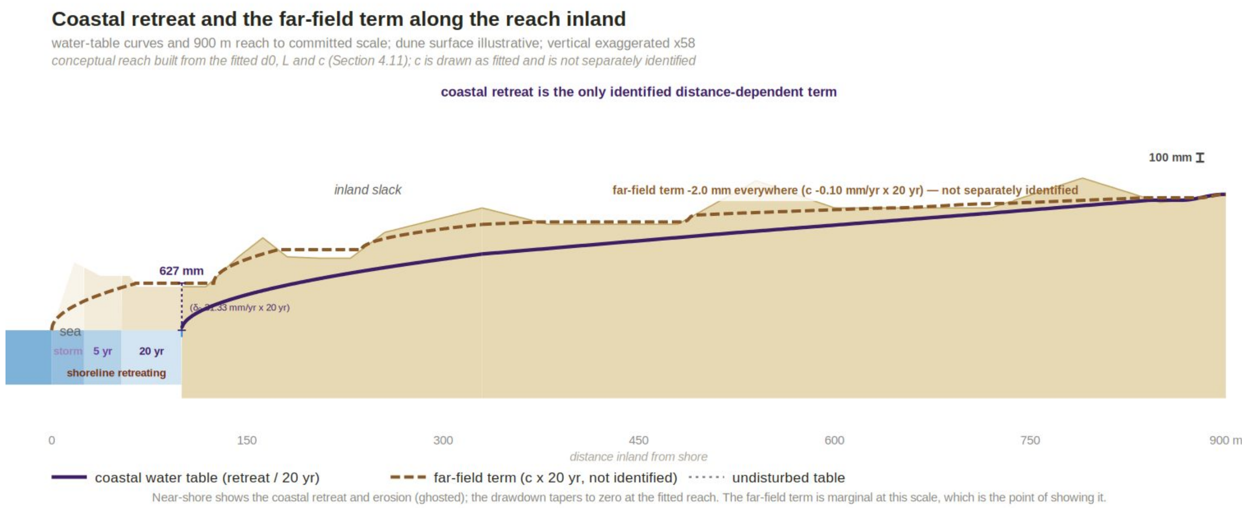

Figure 21. *Conceptual reach of coastal retreat and the far-field term. Long-section from the eroding Caernarfon Bay shoreline inland to 900 m, drawn so that the fitted inland reach L_cg = 895 m (§4.11) falls at the right-hand edge; the dune surface and its slack are illustrative, the vertical scale is exaggerated ×62, and a 100 mm bar is given for reference. The coastal water table (solid) is the fitted coast-edge trend accumulated over a twenty-year horizon, declining linearly to zero at the fitted reach; the undisturbed table (dotted) is the same surface without it. At 150 m, where the headline rate is quoted, the twenty-year accumulation is 524 mm. The curve is drawn in to the shoreline, where the fitted amplitude* δ_0 *= 31.35 mm/yr gives 627 mm; that value is an extrapolation beyond the nearest well, which sits at 147 m. The fitted far-field constant c_far (−0.10 mm/yr, ≈ −2.0 mm accumulated over the same horizon) is a window statistic rather than a rate (§4.11) and is not drawn: over this reach coastal retreat is the only identified distance-dependent term. The ghosted wedges at the shore mark the shoreline positions implied by a single 6 m storm event and by five- and twenty-year accumulations of the mean retreat rate. The section is a schematic built from the fitted parameters rather than a modelled water-table surface. Adapted from the technical report (Hollingham, 2026a).*

This forcing also reframes the C5 Coastal Forest anomaly. C5 shows the steepest summer-minimum decline in the network — several times that of any other cluster — which, given that all clusters share the same climate forcing, argues against a purely climatic explanation. The position analysis (Section 5.2) and the coastal-gradient analysis converge on the same explanation: C5 occupies the lowest, most coast-proximal position in the network, where an inferred thin saturated thickness would amplify the head response to any forcing and where the eroding shoreline lies closest. The single C5 well that enters the gradient subset (nw9, 419 m from the coast) shows a decline of −32.7 mm/yr — the steepest and one of the most significant in the network — and the fitted gradient predicts a coastal-retreat deepening of roughly 15 mm/yr at that position under either functional form. Because nw9 is a forested well it is excluded from the open-dune (forest-free) regression and so functions as an out-of-sample sentinel: its observed decline tests the fitted gradient at a near-coast position without contributing to it. The agreement at that position — an observed −32.7 mm/yr against a fitted coastal contribution of about −15 mm/yr, the residual consistent with the site-wide decline and with the substrate amplification developed in Section 5.2 — is a further corroboration of the gradient at a near-coast position, complementing the shore-normal transect (Section 4.11). A single well cannot, however, carry a quantified cluster attribution, and the C5 figure should be treated as corroborating the gradient's direction and magnitude rather than as an independent estimate; but the convergence of the substrate, position and gradient evidence makes coastal proximity, not canopy, the leading explanation for C5's exceptional trajectory. The companion paper develops the management implication; here the point is characterization — that the framework distinguishes a coast-proximal, structurally amplified decline from the climate background, a distinction that bears directly on how network-wide summer-minimum trends should be attributed.

One boundary mechanism operates alongside the retreat gradient at the most coast-proximal wells: the discharging foreshore is a tidal boundary, so near-coast water tables couple to a constant-head sea boundary that buffers their short-term variability and holds their mean toward the shoreline. This tidal coupling and the multi-year shoreline retreat are complementary rather than competing — the former smooths event-scale fluctuation at a fixed boundary position, the latter migrates that boundary landward over years — and the C5 decline is read against the slower retreat process. The tidal coupling is noted here as the boundary condition that makes near-coast records intrinsically less climate-legible, not as an alternative explanation for the trend.

The decline observed away from the coast — in the clusters that lie beyond the fitted reach and carry no gradient component — is a candidate for further mechanistic work, with at least two non-exclusive climate drivers. What the gradient regression cannot supply is the quantity those drivers would be fitted to: the panel absorbs the spatially uniform forcing through the cumulative-water-balance covariate, and a constant offset trades off exactly against the trend contribution that covariate carries, so no residual far-field rate is isolated here (Section 4.11). The mechanisms below are accordingly motivated by the decline the network records, not by a fitted background, and climate forcing, separately measured over the station record, is the leading candidate for it. The deterioration is concentrated in the summer months, consistent with rising evaporative demand under the documented post-2013 summer warming (Figure 3) acting through the atmospheric-draw term as elevated potential evapotranspiration. That route is, if anything, understated by the forcing: Thornthwaite carries the heat index in its denominator and constructs it from the same temperatures, so the formula partly cancels its own response to warming. Across the station record a 7.2% rise in mean annual temperature (0.73 °C between 1931–1960 and 1996–2025) produced a rise of only 3.0% in mean annual PET, an elasticity of 0.42, so the evaporative demand implied by the temperature record exceeds the demand the model was forced with. A reduction in effective recharge is also plausible — for example a shift in the within-month distribution of rainfall, expressed as a declining recharge sensitivity. The two routes sit differently within the framework: the warming signal enters directly through the model's potential-evapotranspiration forcing, whereas a sub-monthly rainfall redistribution is invisible to a monthly model and would surface only as an apparent decline in recharge sensitivity. The framework makes the network-wide signal visible but does not, on the present analysis, resolve which route dominates, or what share of the observed decline either accounts for; both are noted as forcings the characterization exposes rather than results the paper establishes.

5.6 Predictive skill and transferability

The practical contribution of the framework rests on the benchmarking result. In one-step diagnostic mode the state-space and traditional linear models are nearly indistinguishable, because each prediction is corrected by the previous observation; the distinction emerges under iterative simulation, where the SSM achieves a median Nash–Sutcliffe efficiency of 0.72 against the linear model's −0.03 — a negative efficiency, meaning the linear model leaves more error than the observed long-term mean would — and returns a positive iterative NSE at all but one of the 66 reference wells, against 30 for the linear model. The explicit head-dependent drainage term captures the storage-decay memory that prevents the cumulative drift to which the linear model is prone. The spatial pattern of the gain is itself a diagnostic: the largest improvements concentrate in the lake-proximal eastern aquifer where head-dependent drainage is strongest, and the smallest occur in the forest interior where weak drainage makes the storage term nearly dispensable. The improvement map therefore delineates which parts of the aquifer are governed by nonlinear storage dynamics and which approximate a linear climate response — a structural characterization obtained without any invasive measurement.

This is the basis for transferability. The framework requires only three fitted parameters per well, a uniform drainage datum, and monthly rainfall and temperature from a single climate station. It does not require hydraulic-conductivity measurements, aquifer-thickness surfaces, or lateral-flow modelling — precisely the parameter-sparsity constraints that produced calibration difficulties for an earlier distributed-model effort at this site (Betson et al., 2002). The displacement formulation is the methodological choice that enables this: by referencing drainage to a fixed sub-surface datum rather than to the ground surface, it captures the depth-dependence of the drainage response without site-specific calibration of the drainage geometry, and it resolves a physically meaningful drainage coefficient even in the low-gradient forest interior. Any coastal dune aquifer with a manual dipwell network and a nearby climate station can be parameterized in the same way, and the diagnostic syntheses demonstrated here — the **t½/ΔNSE/Sy** characterization, the coefficient atlas, the residual field — follow directly from that parameterization. One supplementary step reaches outside this core: rendering the forest-interception drawdown as a physical reach in metres (Figure 19) requires a hydraulic diffusivity, for which a single literature conductivity (K ≈ 6 m/day; Betson et al., 2002) and a nominal saturated thickness are assumed — the latter uncertain by a factor of two or more. These enter only that illustrative reach length; the core parameterization, the diagnostic syntheses above, and the empirically fitted coastal reach (Figure 20) all remain free of conductivity and thickness inputs. The forest reach is accordingly an order-of-magnitude diagnostic rather than a calibrated prediction.

5.7 Framework limitations

The framework's strengths and its limits are two sides of the same design choice. Because it fits a lumped mass balance independently at each well and interpolates the results geometrically, it does not resolve lateral groundwater flow. The interpolated coefficient and head surfaces are aggregators of point responses, not a physical flow model, and cannot represent cluster-to-cluster hydraulic coupling, the lateral redistribution of localized recharge perturbations, or the diffusive response of the aquifer to a change in boundary flux. The affinity analysis and the Darcy field show that the clusters are connected along a continuous downslope drainage pathway, so whether a signal originating in one part of the aquifer propagates to another at magnitudes below the framework's resolution cannot be answered here; that is a question for a calibrated continuous-flow model. The critical unknowns for such a model are the very quantities the present framework avoids — hydraulic conductivity, currently constrained by a single tracer test (6.0 m/day; Betson et al., 2002), and aquifer thickness, inferred from a handful of borehole logs across the site. A measured K would also tighten the only K-conditioned output of the present framework — the forest-drawdown reach length in Figure 19 — but, as noted in Section 5.6, all other framework outputs (SSM coefficients, WTF specific yields, substrate-gradient interpretation, coastal-retreat regression and residual field) are K-independent and would not change with a measured K. Slug tests at representative wells per cluster and a ground-penetrating-radar survey of aquifer thickness are the priority field measurements to enable a flow model, and would also support an independent vadose-zone test of the canopy-driven recharge-suppression signal. The same campaign would test the structural interpretation of the C3 transition (Section 5.1): a south-west to north-east transect resolving the depth to the estuarine-clay base, carrying the Caernarfon Bay coastal-retreat rate as a covariate, would establish whether the eastward-to-south-westward buffering gradient reflects a single dipping clay base — one thickening sand column — rather than two distinct substrate blocks, and would separate that structural control from the confounded coastal and plantation-margin influences.

Several further limitations qualify specific results. The monthly measurement resolution cannot isolate individual storm events, so the water-table-fluctuation specific yields conflate gravity drainage with capillary-fringe release and are upper bounds rather than point estimates (Healy and Cook, 2002; Scanlon et al., 2002). Thornthwaite PET is a temperature-and-daylength index whose absolute magnitude is uncertain; the framework's internal consistency does not depend on it, but absolute flux estimates derived from it do. The coastal-retreat regression is a steady-state spatial model: it resolves the first-order distance gradient but not the multi-year transient over which an erosion signal propagates inland, and direct measurement of present-day retreat rates at the south-western dune front remains a monitoring priority. The water-balance residual field is spatially structured and reproducible but, as discussed, its attribution between a genuine lateral flux and a modelling artefact is unresolved. Finally, the close correspondence between DEM-derived drainage pathways and the cluster boundaries supports the use of surface topography as an operational proxy for subsurface flow divides, but the two are co-determined by the underlying till and bedrock architecture rather than one causing the other; the proxy is reliable for routing in the absence of high-resolution bedrock mapping, not a substitute for it.

6. Conclusions

This study sets out a parameter-sparse state-space framework for characterizing the hydrogeological architecture of a coastal dune aquifer from routine monitoring data, and demonstrates it across 88 dipwells and 21 years of monthly records at Newborough Warren. The principal conclusions are:

1. **Five hydrogeological clusters govern site hydrology.** The Pearson affinity audit of hydrograph similarity independently confirms cluster assignment

for 95% of reference-network wells (Section 4.3), and the cluster-level contrasts in SSM coefficients, drainage half-life and specific yield provide mechanistic substantiation. The five units resolve as — a responsive eastern block (C1 Lake Edge, C2 Dune), a buffered Western Residual (C3), and two forested clusters distinguished by topographic position and inferred substrate (C4 Main Forest, high atmospheric draw and weak drainage on the elevated ridge flank; C5 Coastal Forest, on the lower coastal ground). Within the forest zone, elevation alone explains 97% of the variance in atmospheric draw, identifying position and elevation rather than canopy as the dominant within-forest control.

2. **The aquifer's storage varies smoothly, without discontinuity at the cluster margins; the cluster contrasts are surface-mediated.** Once corrected for canopy interception, the C4 forest specific yield falls within the open-dune range and the interpolated specific-yield surface shows no discontinuity at the cluster margins; the C5 estimate is consistent with this but is the least reliable in the network and is treated as weakly corroborative. The forest clusters' distinctive coefficient signature therefore derives from a surface boundary condition — canopy interception suppressing recharge — rather than from a different subsurface architecture.

3. **The displacement-formulation SSM closes the water balance and substantially improves forecast skill.** The three-term model closes the monthly water balance to within 1.8% of total losses at every cluster, and achieves a median iterative Nash–Sutcliffe efficiency of 0.72 against −0.03 for a traditional linear model — a negative efficiency, meaning the linear model performs worse than predicting the long-term mean — with a positive iterative NSE at 65 of 66 wells against 30 of 66. The spatial pattern of the improvement is itself a non-invasive diagnostic of where head-dependent drainage memory governs the aquifer.

4. **The water balance closes on average and its residual field is spatially uniform.** Residuals lie within ±0.01 m/month at 60 of the 66 reference wells and carry no gradient across the site; three open-dune wells exceed +0.02 m/month, no forest or ridge-proximal well does, and no unmodelled lateral flux is required to close the balance at any cluster. A cross-correlation lag test and a seasonal climatology test on the monthly residual likewise return null results for ridge-derived input, though the monthly sampling interval is coarse relative to the transit times expected of fracture flow and the nulls

therefore bound what this record can resolve rather than excluding the mechanism. The residual field is a structural diagnostic of model adequacy, not a quantified flux map.

5. **Coastal retreat is a distinct, spatially structured forcing on long-term water-table trends.** A network-scale gradient regression resolves a coast-edge deepening of roughly 26 mm/yr at 150 m from the shoreline, declining over an inland reach of order 1 km, accounting for about a quarter of the observed summer-minimum decline at the near-coast Western Residual, and is corroborated by the progressive deepening of the coast-to-inland head difference along a shore-normal transect; it is spatially structured — strongest at the shore and decaying to nothing over that reach — rather than a uniform decline. The exceptional decline of the coast-proximal C5 cluster is best explained by coastal position and thin saturated thickness rather than by its canopy. Distinguishing this coastal forcing from the climate background is necessary for any correct attribution of network-wide summer-minimum trends.

6. **The framework is transferable.** It characterizes aquifer architecture — storage, recharge sensitivity, drainage memory and flow structure — from manual dipwell records and a single climate station, without coring, hydraulic testing or calibrated continuous-flow modelling. Its principal limitation is the converse of this strength: as a geometric aggregator of point responses it cannot resolve cluster-to-cluster hydraulic coupling, for which a calibrated flow model, and the hydraulic-conductivity and aquifer-thickness measurements it requires, would be needed.

The hydrological consequences of the dune-scraping and plantation-clearfell interventions undertaken at the site are quantified within a before-after-control-impact framework in the companion paper (Hollingham, 2026b).

Source data

For review only — every figure and table below traces to a committed file in the project repository (paths relative to outputs/, or to config.py / data/ where noted). This block lets reviewers verify each value against its producing artefact and is to be removed at production.

Item	Committed source
Table 1 — cluster mechanistic characterization	03_state_space_model/ 03_03_cluster_mechanistic_coefficients.csv
Table 2 — mean monthly head-space water balance	16_water_balance/16_water_bal_table.csv
Table 3 — annual volumetric water balance	16_water_balance/16_water_bal_vol_table.csv
Table 4 — specific yield by cluster (WTF method)	17_wtf_specific_yield/17_wtf_01_sy_estimates.csv
Table 5 — SSM vs traditional-linear-model benchmarking	08_model_benchmarking/ 08_lcsc_04_table3_benchmark_summary.csv
Table 6 — per-well SSM coefficient ranges by cluster	07_spatial_coefficients/07_coeff_05_cluster_ranges.csv
Table 7 — drainage decay half-life $t½ = \ln(2)/\beta_3$ by cluster	18_wtf_spatial/18_wtf_05_storage_drainage_index.csv
Table 8 — within-forest spatial predictors of SSM coefficients	10c_forest_zone_correlations.csv
Table 9 — decomposition of cluster summer-minimum decline	25_coastal_gradient/25_03_cluster_partition.csv
Figure 1 — site topography, geology and monitoring network	01_locations.csv (+ DEM/KML in data/geo/)
Figure 2 — climate forcing over the monitoring period	01_climate.csv
Figure 3 — summer-temperature record	00_climate_summary/ 00_03_summer_warming_stats.csv
Figure 4 — Ward's dendrogram	02_clustering/02_07_cluster_membership_k5.csv
Figure 5 — spatial cluster distribution (Pearson affinity)	05_pear_membership_audit.csv (+ 01_locations.csv)
Figure 6 — cluster-centroid hydrographs	02_clustering/02_09_cluster_amplitude_summary.csv

and amplitudes	
Figure 7 — water-balance decomposition by cluster	16_water_balance/16_water_bal_table.csv
Figure 8 — WTF specific-yield surface	18_wtf_spatial/18_wtf_01_well_sy_estimates.csv
Figure 9 — SSM-over-TLM Nash–Sutcliffe improvement	08_model_benchmarking/08_perwell_nse.csv
Figure 10 — optimal drainage datum and R^2 gain	03_state_space_model/ 03_09_well_optimal_datums.csv; 03_08_datum_sensitivity.csv
Figure 11 — β_1 recharge-sensitivity surface	07_spatial_coefficients/07_coeff_maps_data.csv
Figure 12 — β_2 atmospheric-draw surface	07_spatial_coefficients/07_coeff_maps_data.csv
Figure 13 — β_3 drainage-rate surface	07_spatial_coefficients/07_coeff_maps_data.csv
Figure 14 — drainage decay half-life t½	18_wtf_spatial/18_wtf_05_storage_drainage_index.csv
Figure 15 — aquifer diagnostic synthesis	18_wtf_spatial/18_wtf_05_storage_drainage_index.csv; 08_model_benchmarking/08_perwell_nse.csv; 18_wtf_spatial/18_wtf_01_well_sy_estimates.csv
Figure 16 — mean annual water-table surface with Darcy flow-directions	20_spatial_figures/20_head_surface_streams.png (head surface IDW-interpolated from per-well mean annual head in 03_master_data.csv; vectors are direction-only)
Figure 17 — SSM water-balance residual field	20_spatial_figures/20_residual_perwell.csv
Figure 18 — coastal-retreat gradient (a, b, c)	25_coastal_gradient/25_report_numbers.csv; 25_coastal_gradient/25_03_cluster_partition.csv; 38_coastal_transect/38_results.txt
Figure 19 — illustrative forest reach	20_drawdown_propagation_nohead.png; schematic, parameters from config.py (FOREST_INTERCEPTION, DRAWDOWN_H0/K/B), rendered via Script 20
Figure 20 — episodic coastal reach	schematic; parameters from 25_coastal_gradient/25_report_numbers.csv and config.py (COAST_RETREAT_*)
Figure 21 — spatial reach and development timescale of interventions and coastal retreat	09g_coastal_vs_climate_reach.png; schematic, δ_0/L_cg from 25_coastal_gradient/25_01_panel_fit_parameters.csv, rendered via Script 09g

References

Akaike, H. (1974) A new look at the statistical model identification. *IEEE Transactions on Automatic Control* 19(6), 716–723. doi:10.1109/TAC.1974.1100705

Bear, J. (1972) *Dynamics of Fluids in Porous Media.* New York: American Elsevier.

Betson, M., Connell, L. and Bristow, C. (2002) Groundwater Modelling of Newborough Warren. In: Bristow, C. (ed.) *The Impact of Forestry on Coastal Geomorphology at Newborough Warren/Ynys Llanddwyn NNR, SSSI, pSAC. Volume 5: Hydrogeology.* Report to the Countryside Council for Wales, Contract FC 73-05-18. Department of Geological Sciences, UCL, and School of Earth Sciences, Birkbeck, University of London, July 2002.

Curreli, A., Wallace, H., Freeman, C., Hollingham, M., Stratford, C., Johnson, H. and Jones, L. (2013) Eco-hydrological requirements of dune slack vegetation and the implications of climate change. *Science of the Total Environment* 443, 910–919. doi:10.1016/j.scitotenv.2012.11.035

Davy, A.J., Grootjans, A.P., Hiscock, K. and Petersen, J. (2006) *Development of Eco-Hydrological Guidelines for Dune Habitats — Phase 1.* English Nature Research Reports No. 696. Peterborough: English Nature.

Fetter, C.W. (2001) *Applied Hydrogeology* (4th ed.). Upper Saddle River, NJ: Prentice Hall.

Forgrave, A. (2020) Dramatic pictures show the beautiful little corner of North Wales that's fast being lost to the sea. *North Wales Live / Daily Post,* 22 January 2020.

Freeman, S. (2008) *The hydrological impact of Corsican pine plantation on the dune system at Newborough Warren, Anglesey.* Unpublished BSc Honours dissertation, University of Wales, Bangor.

Freeze, R.A. and Cherry, J.A. (1979) *Groundwater.* Englewood Cliffs, NJ: Prentice-Hall.

Gerla, P.J. (2019) Monitoring and modeling the effect of agricultural drainage and recent channel incision on adjacent groundwater-dependent ecosystems. *Water* 11(4), 863. doi:10.3390/w11040863

Healy, R.W. and Cook, P.G. (2002) Using groundwater levels to estimate recharge. *Hydrogeology Journal* 10(1), 91–109. doi:10.1007/s10040-001-0178-0

Hollingham, M. (2026a) Hydrogeological Dynamics, Behavioural Clustering and Management Intervention Analysis at Newborough Warren Coastal Sand Dune Aquifer, Wales. Technical report, June 2026. https://newbroman.github.io/Newborough_Hydrology/ Hollingham, M.

(2026b) Hydrological response of a coastal dune aquifer to plantation clearfell and dune scraping: a BACI analysis of a 21-year manual record. *In preparation.*

Knotters, M. and van Walsum, P.E.V. (1997) Estimating fluctuation quantities from time series of water-table depths using models with a stochastic component. *Journal of Hydrology* 197(1–4), 25–46. doi:10.1016/S0022-1694(96)03278-7

Pye, K. and Blott, S.J. (2024) *Dune rejuvenation AfterLIFE options for Twyni Penrhos, Newborough.* NRW Evidence Report No. 787, 73 pp. Bangor: Natural Resources Wales.

Ranwell, D.S. (1958) Movement of vegetated sand dunes at Newborough Warren, Anglesey. *Journal of Ecology* 46, 83–100.

Rao, A.R. and Srinivas, V.V. (2006) Regionalization of watersheds by fuzzy cluster analysis. *Journal of Hydrology* 318(1–4), 57–79.

Rhind, P.M., Blackstock, T.H., Hardy, H.S., Jones, R.E. and Sandison, W. (2001) The evolution of Newborough Warren dune system with particular reference to the past four decades. In: Houston, J.A., Edmondson, S.E. and Rooney, P.J. (eds) *Coastal Dune Management.* Liverpool: Liverpool University Press, pp. 345–379.

Robins, N.S., Pye, K. and Wallace, H. (2013) Dynamic coastal dune spit: the impact of morphological change on dune slacks at Whiteford Burrows, South Wales, UK. Journal of Coastal Conservation 17(3), 473–482. doi:10.1007/s11852-013-0245-4

Rousseeuw, P.J. (1987) Silhouettes: a graphical aid to the interpretation and validation of cluster analysis. *Journal of Computational and Applied Mathematics* 20(1), 53–65. doi:10.1016/0377-0427(87)90125-7

Scanlon, B.R., Healy, R.W. and Cook, P.G. (2002) Choosing appropriate techniques for quantifying groundwater recharge. *Hydrogeology Journal* 10(1), 18–39. doi:10.1007/s10040-001-0176-2

Stratford, C., Ratcliffe, J., Hughes, A.G., Roberts, J. and Robins, N.S. (2007) Complex interaction between shallow groundwater and changing woodland, surface water, grazing and other influences in partly wooded duneland in Anglesey, Wales. In: Ribeiro, L., Chambel, A. and Condesso de Melo, M.T. (eds) *IAH 35th Congress, Groundwater and Ecosystems,* Lisbon.

Ward, J.H. (1963) Hierarchical grouping to optimize an objective function. *Journal of the American Statistical Association* 58(301), 236–244.

Young, P.C. (2011) *Recursive Estimation and Time-Series Analysis: An Introduction for the Student and Practitioner.* Berlin: Springer.